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IN THEIR OWN WORDS: REASONING TRACES TAILORED FOR SMALL MODELS MAKE THEM BETTER REASONERS

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ABSTRACT

Stronger learning signals do not necessarily produce a stronger small reasoning model. Off-policy supervised fine-tuning (SFT) supplies stronger traces that may be incompatible with the student. On-policy post-training instead remains constrained by the quality of the student's own trajectories: reinforcement learning (RL) provides little learning signal because a small student rarely produces a correct rollout. Consequently, effective post-training must supply stronger reasoning in a form the student can learn from. We hypothesize that a small number of tokens with very low probability under the student can make an otherwise strong reasoning trace difficult to learn. To test this hypothesis, we introduce Interleaved-Policy Distillation (IPD), which generates SFT data balancing teacher guidance and student compatibility. During generation, the student replaces teacher proposals that have low probability under its own policy, allowing the teacher to continue reasoning from a prefix the student can support. IPD improves reasoning across model families, datasets, and varying degrees of student intervention; for example, it raises the seven-benchmark average from 24.55 to 29.72 on Qwen3-0.6B/s1K. It outperforms baselines with comparable training budgets and broadens problem-solving coverage, while ablations show that filtering traces or masking losses cannot match the gains from revising trajectories during generation. Analyses link transfer failures to the rare tokens the student finds least likely and show that the source of the tokens also matters. Small models reason better when they learn strong solutions in a form they can reach in their own words.

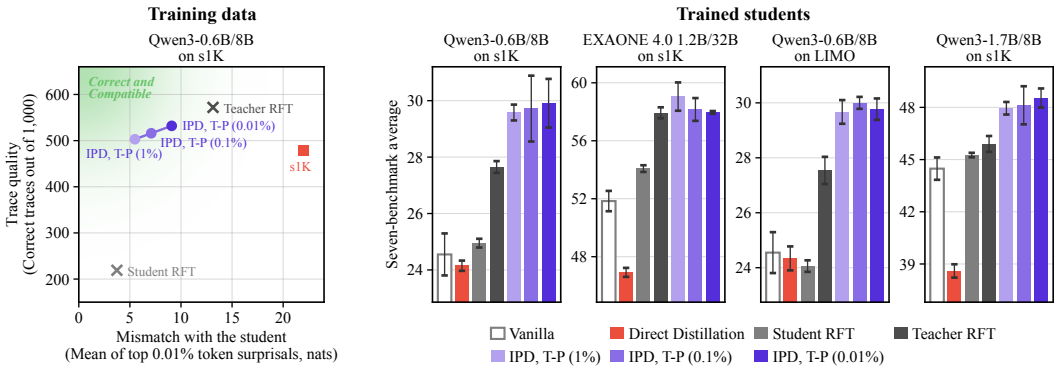


Figure 1: Training data in the primary setting and student performance across four settings. All IPD results shown use the teacher-propose (T-P) variant. *Left:* Each point is one method's training set, plotted by verified correct traces and the mean of the top 0.01% token surprisals under the student. Student RFT is compatible but teaches little; the original s1K traces used for Direct Distillation are strong but incompatible. *Right:* Direct Distillation degrades the student below Vanilla, while IPD outperforms every other method in seven-benchmark average accuracy. Error bars show one standard deviation across three training seeds. **Takeaway:** correct and compatible data trains the best student.

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1 INTRODUCTION

Stronger learning signals do not necessarily produce a stronger small reasoning model. Off-policy supervised fine-tuning (SFT) provides stronger traces that may be incompatible with the student: fine-tuning Phi-4-Mini on s1K (Muennighoff et al., 2025) reduced AIME24 accuracy from 10.0 to 3.0 and MATH500 accuracy from 71.8 to 47.0 (Xu et al., 2025a). On-policy methods instead remain constrained by the quality of the student's own rollouts. Reinforcement learning (RL) provides little learning signal because a small student rarely produces a correct rollout. On-policy distillation (OPD) supplies dense teacher feedback, but student rollouts can remain far from the teacher's reasoning distribution, forcing the teacher to condition on low-quality prefixes that weaken its guidance (Zhu et al., 2026). Effective post-training must therefore supply stronger reasoning in a form the student can learn from.

We hypothesize that a small number of tokens with very low student probability can make otherwise strong reasoning difficult to learn. A trace can appear compatible on average even when it contains a few tokens the student struggles to follow. Only a small fraction of tokens steer the reasoning that follows (Wang et al., 2025), so once a problematic token enters the prefix, the rest of the solution builds on a choice the student would rarely make. These problematic tokens cannot be treated in isolation because their effects are embedded in subsequent reasoning. The challenge is to preserve strong reasoning while making both the trajectory and its learning targets compatible with the student.

To test this hypothesis, we introduce Interleaved-Policy Distillation (IPD), in which a teacher and student jointly construct each reasoning trace. In teacher-propose IPD, the teacher proposes each token, the student replaces proposals whose probability under its own policy falls below a threshold τ, and the teacher continues from the revised prefix. IPD trains on the resulting interleaved traces, making the student a source of supervision as well as conditioning prefixes. Reversing the roles gives student-propose IPD, a control that tests whether the proposal direction drives the gains (Figure 2).

Strong reasoning transfers more effectively when its trajectories are adapted to the student. Across four settings spanning Qwen3 and EXAONE students (Yang et al., 2025; Bae et al., 2025) with 0.6B–1.7B parameters and the s1K and LIMO datasets (Muennighoff et al., 2025; Ye et al., 2025), Direct Distillation leaves the student below its initial performance. Teacher-propose IPD consistently improves performance and achieves the highest seven-benchmark average in every setting, outperforming both Student RFT and Teacher RFT (Yuan et al., 2023) with only 1,000 s1K or 800 LIMO training questions. For example, on Qwen3-0.6B/s1K, IPD raises the average from 24.55 to 29.72.

This advantage persists across two orders of magnitude in τ, with teacher-propose IPD exceeding Teacher RFT in all twelve setting–threshold pairs. In the primary setting, it also outperforms RL and OPD under comparable training budgets. Increasing Student RFT from 8 to 100 attempts more than doubles its correct traces but leaves performance unchanged. Ablations show that filtering traces, masking token losses, mixing teacher and student distributions, and matching teacher logits on interleaved prefixes fall short of teacher-propose IPD. These comparisons support selectively revising trajectories during generation and learning from the jointly generated tokens.

IPD reduces the rare, severe mismatches that motivate its interleaving rule. We quantify these through the mean of the top 0.01% token surprisals under the initial student. In the primary setting, teacher-propose IPD reduces this mean from 21.99 nats for Direct Distillation on s1K to 5.50–9.09 nats (Figure 1). Mean surprisal is a weaker guide: student-propose IPD yields lower mean surprisal than teacher-propose IPD, yet trains weaker students. Surprisal alone, however, does not fully explain transfer: the source of the reasoning tokens also matters. Even at a comparable top-0.01% mean, teacher-propose IPD outperforms student-propose IPD. This gap suggests that compatibility and token source jointly shape the value of supervision.

IPD also expands the range of problems the student can solve on its own. At the largest matched sampling budget, it leads Vanilla and Direct Distillation on Pass@k in 27 of 28 benchmark–setting pairs. Whereas average accuracy measures how often a sampled solution is correct, Pass@k measures whether any of k rollouts is. This is the quantity that limits RL, which learns from a problem only when the student produces a correct rollout. Broader coverage therefore motivates IPD as an initialization for subsequent on-policy learning, giving RL more successes to reinforce and OPD more useful contexts for teacher feedback. **Small models reason better when they learn strong solutions in a form that they can reach in their own words.**

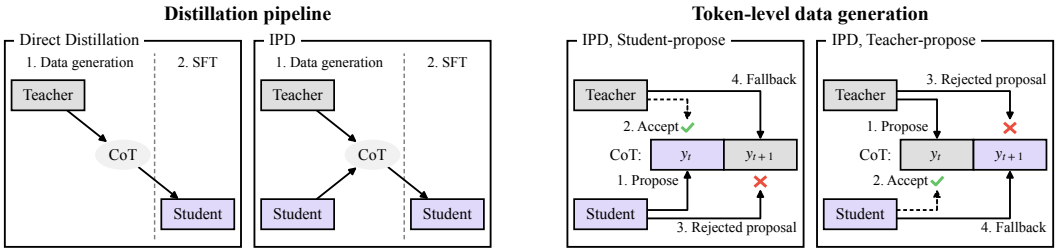


Figure 2: *Left:* Direct Distillation trains on fixed teacher traces; IPD constructs each trace from both policies. *Right:* a proposal is accepted at t and rejected at $t+1$, where the other policy supplies a fallback token. Both tokens enter the shared prefix and SFT targets.

2 Related Work

Student-aligned distillation. Teacher traces differ in how easily a given student can learn from them. Prior work ranks teachers using statistics of student gradients (Panigrahi et al., 2026), selects responses using student likelihood (Zhang et al., 2025a) or rank-to-surprisal ratios (Yang et al., 2026b), and contrasts teacher- and student-generated responses during training (Ko et al., 2025). Other approaches score responses using student likelihood within local contexts (Just et al., 2026), train on teacher prefixes the student judges sufficient and then completes (Liu et al., 2026b), or trace teacher-originated sentences across models to select data (Liu et al., 2026a).

Reshaping teacher guidance. OPD provides teacher feedback on student rollouts (Agarwal et al., 2024), but their prefixes can weaken that guidance (Zhu et al., 2026). Making teaching more useful to the student can involve adapting teacher prompts and policies (Kim et al., 2024; Cetin et al., 2025), optimizing the teacher with DPO on student reference-loss changes (Li et al., 2025), or designing data and curricula using knowledge probes (He et al., 2026a). Teacher guidance can also structure the traces themselves, through expert prefixes that anchor RL rollouts (Zhang et al., 2025b) or explicit reasoning-step roles (Wen et al., 2026). The student can participate directly by selecting teacher continuations using perplexity (He et al., 2026b), or completing and rewriting teacher-started solutions (Yang et al., 2026a). Other work combines teacher ranking of student-proposed token candidates with distribution matching weighted by student–teacher entropy gaps (Kim & Baek, 2026).

These methods select or construct traces over whole responses or multi-token continuations, or apply token-level feedback to sequences the student has already generated. IPD instead uses token-level acceptance and replacement during generation and trains directly on the resulting tokens. Speculative Knowledge Distillation (SKD) also interleaves student proposals and teacher corrections, but matches the teacher distribution at each position (Xu et al., 2025b), so student tokens serve only as conditioning prefixes. Our primary configuration, teacher-propose IPD, also reverses SKD's proposal direction: the teacher proposes, and the student replaces rejected proposals. Our control, student-propose IPD, shares SKD's proposal direction but retains IPD's jointly generated SFT targets.

3 Method

3.1 Student Support

Let x denote a problem, $y = (y_1, \ldots, y_L)$ a reasoning trace, and p_T and p_S the teacher and initial student policies. The models share a tokenizer so that each can score the other's token proposals. Given a prefix $y_{<t}$, a candidate next token z has student support at threshold τ when $p_S(z \mid x, y_{<t}) \geq \tau$. Its student surprisal is $-\log p_S(z \mid x, y_{<t})$, measured in nats. For scale, 4.61 nats corresponds to a student probability of 1%, and 20 nats to about 2×10^{-9}.

To assess training datasets, we use *upper-tail mean surprisal*: the mean of the top 1%, 0.1%, or 0.01% of a dataset's token surprisals under the student. These means reveal rare mismatches that an average over all tokens can obscure, with smaller fractions isolating more extreme mismatches. The top-0.01% mean covers the few tokens the student finds least likely, which our analyses link to transfer failures; we call it the *extreme-tail mean surprisal*.

3.2 TEACHER-PROPOSE IPD

Our primary configuration, teacher-propose IPD (T-P), lets the teacher lead generation and the student intervene at proposals it finds unlikely. Both policies condition on the shared prefix $y_{<t}$. The teacher samples a token z, which is accepted if $p_S(z \mid x, y_{<t}) \geq \tau$. Otherwise, the student samples a replacement, or fallback token, from its own policy without a second support test. Both policies remain frozen during generation.

Algorithm 1 expresses this rule using a proposer p_P and a gate p_G, which scores proposals and supplies replacements. T-P sets $p_P = p_T$ and $p_G = p_S$; student-propose IPD (S-P) reverses these roles. T-P thus adapts teacher reasoning to the student, whereas S-P corrects student reasoning with teacher tokens. Because each committed token extends the shared prefix, a replacement also redirects the proposals that follow.

Algorithm 1 Interleaved-policy rollout of one trace

Require: problem x, proposer p_P, gate p_G, threshold τ, budget L
for $t = 1, \ldots, L$ **do**
  $z \sim p_P(\cdot \mid x, y_{<t})$
  if $p_G(z \mid x, y_{<t}) < \tau$ **then**
    $z \sim p_G(\cdot \mid x, y_{<t})$
  end if
  $y_t \leftarrow z$
  if y_t terminates **then break**
end for
return y

We then fine-tune the student on the resulting traces with ordinary SFT. Every committed token, including each student replacement, is therefore an SFT target.

At $\tau = 0$, T-P accepts every proposal and reduces to teacher-only generation; at the other extreme, rejecting every proposal reduces it to student-only generation. At intermediate thresholds, accepted teacher proposals satisfy

$$-\log p_S(y_t \mid x, y_{<t}) \leq \log(1/\tau). \tag{1}$$

The ceilings are 4.61, 6.91, and 9.21 nats for $\tau = 1\%, 0.1\%, 0.01\%$. Student replacements are not gated, so a dataset's upper-tail mean surprisal can exceed these ceilings. The threshold therefore controls how readily the student replaces teacher proposals and redirects subsequent reasoning.

4 IPD ROBUSTLY IMPROVES REASONING DISTILLATION

4.1 CONSISTENT GAINS ACROSS MODELS AND DATASETS

Teacher-propose IPD achieves the highest seven-benchmark average across all four model–dataset settings. Table 1 covers Qwen3-0.6B/8B on s1K and LIMO, EXAONE 1.2B/32B on s1K, and Qwen3-1.7B/8B on s1K, with model pairs written as student/teacher. Its baselines span the trade-off between strong reasoning and student compatibility. Direct Distillation uses the released traces of s1K-1.1 and LIMO-v2 (s1K and LIMO hereafter). Student RFT and Teacher RFT use correctness-verified traces sampled for the same questions from the student and the teacher, respectively. Vanilla is the initial student. Percentages in IPD labels denote the acceptance threshold τ, and we use $\tau = 0.1\%$ as the representative configuration.

Within each setting, all methods use the same generation budget, SFT schedule, and evaluation, so they differ only in their training data. Each data-generation method has a budget of eight attempts per question. Unsolved IPD questions contribute partial traces of at most 128 tokens generated by the same IPD condition (Ji et al., 2025). We select the SFT schedule through an independent held-out hyperparameter search for each setting (Appendix A.3), yielding 15 epochs at 3×10^{-5} for the Qwen3 settings and 30 epochs at 10^{-5} for EXAONE. Evaluation uses Avg@8 on AIME24/25/26 and HMMT26, and Avg@4 on MATH500, OlympiadBench, and GPQA-D (Mathematical Association of America, n.d.; Harvard–MIT Mathematics Tournament, 2026; Lightman et al., 2024; He et al., 2024; Rein et al., 2024), averaged over three training seeds.

The results order these methods by how well their data combine strength with compatibility. Direct Distillation, whose traces are the least compatible, falls below Vanilla in all four settings; Student RFT gains less than Teacher RFT, and Teacher RFT less than T-P, in every setting. Neither strength nor compatibility is enough alone. In our primary setting, Qwen3-0.6B/8B on s1K, T-P at $\tau = 0.1\%$ raises the average from 24.55 to 29.72, compared with 24.15 for Direct Distillation and 27.65 for Teacher RFT. Across settings it improves on Vanilla by 3.64–6.32 points, and the gain is broad: it holds in all 28 benchmark–setting pairs. **Adapting teacher reasoning to the student yields consistent gains across the tested models and datasets.**

Table 1: Accuracy (%) on seven benchmarks in four settings. Entries are mean ± sample standard deviation across three seeds; each seed sets training and evaluation, or only evaluation for Vanilla. Average is the unweighted benchmark mean, and IPD percentages are τ. Blue and red shading mark gains and losses relative to Vanilla; bold marks the best value per column and setting.

Method	AIME24	AIME25	AIME26	HMMT26	MATH500	Olympiad	GPQA-D	Average
Qwen3-0.6B/8B on s1K								
Vanilla	$6.25_{\pm 1.67}$	$15.00_{\pm 1.91}$	$10.42_{\pm 1.10}$	$8.96_{\pm 0.79}$	$71.07_{\pm 0.40}$	$37.82_{\pm 0.26}$	$22.35_{\pm 0.91}$	24.55
Direct Distillation	$4.86_{\pm 0.24}$	$18.33_{\pm 2.20}$	$10.14_{\pm 0.64}$	$8.46_{\pm 1.22}$	$71.63_{\pm 0.75}$	$35.87_{\pm 0.27}$	$19.78_{\pm 0.07}$	24.15
Student RFT	$6.11_{\pm 1.34}$	$14.44_{\pm 0.64}$	$10.28_{\pm 0.48}$	$7.70_{\pm 0.95}$	$71.42_{\pm 1.35}$	$38.24_{\pm 0.27}$	$26.43_{\pm 1.07}$	24.95
Teacher RFT	$7.50_{\pm 1.91}$	$19.86_{\pm 0.64}$	$11.11_{\pm 3.07}$	$11.87_{\pm 2.22}$	$74.40_{\pm 0.40}$	$41.31_{\pm 0.25}$	$27.48_{\pm 0.65}$	27.65
IPD, S-P (1%)	$\mathbf{11.25}_{\pm 0.42}$	$19.58_{\pm 0.00}$	$\mathbf{16.11}_{\pm 2.71}$	$12.37_{\pm 2.19}$	$76.45_{\pm 1.55}$	$43.13_{\pm 0.41}$	$24.79_{\pm 1.03}$	29.10
IPD, S-P (0.1%)	$9.44_{\pm 3.01}$	$19.86_{\pm 0.24}$	$14.17_{\pm 0.72}$	$13.01_{\pm 0.44}$	$76.02_{\pm 0.98}$	$41.59_{\pm 0.70}$	$24.28_{\pm 0.65}$	28.34
IPD, S-P (0.01%)	$9.31_{\pm 1.05}$	$18.61_{\pm 1.46}$	$15.56_{\pm 1.58}$	$10.98_{\pm 1.00}$	$74.77_{\pm 0.50}$	$41.86_{\pm 0.60}$	$30.05_{\pm 1.98}$	28.73
IPD, T-P (1%)	$10.28_{\pm 2.44}$	$20.14_{\pm 2.10}$	$12.08_{\pm 1.10}$	$13.64_{\pm 1.65}$	$\mathbf{77.78}_{\pm 0.35}$	$\mathbf{43.18}_{\pm 0.71}$	$29.97_{\pm 1.50}$	29.58
IPD, T-P (0.1%)	$10.14_{\pm 3.15}$	$\mathbf{20.56}_{\pm 0.48}$	$14.03_{\pm 2.77}$	$12.37_{\pm 1.09}$	$76.30_{\pm 1.06}$	$42.89_{\pm 0.85}$	$31.78_{\pm 1.39}$	29.72
IPD, T-P (0.01%)	$8.19_{\pm 1.05}$	$19.86_{\pm 2.96}$	$13.19_{\pm 1.46}$	$\mathbf{14.14}_{\pm 2.87}$	$75.43_{\pm 0.50}$	$41.41_{\pm 0.09}$	$\mathbf{37.16}_{\pm 1.58}$	**29.91**
EXAONE 4.0 1.2B/32B on s1K								
Vanilla	$38.47_{\pm 1.34}$	$41.11_{\pm 3.49}$	$49.58_{\pm 2.92}$	$30.18_{\pm 2.74}$	$89.82_{\pm 0.20}$	$66.01_{\pm 1.15}$	$47.69_{\pm 1.19}$	51.84
Direct Distillation	$33.19_{\pm 1.68}$	$36.94_{\pm 1.97}$	$44.03_{\pm 0.24}$	$31.69_{\pm 0.87}$	$89.77_{\pm 0.32}$	$58.86_{\pm 0.06}$	$33.92_{\pm 0.70}$	46.92
Student RFT	$40.42_{\pm 1.50}$	$44.44_{\pm 1.68}$	$53.75_{\pm 0.42}$	$32.95_{\pm 0.38}$	$90.48_{\pm 0.43}$	$67.68_{\pm 0.72}$	$48.82_{\pm 2.27}$	54.08
Teacher RFT	$\mathbf{48.47}_{\pm 0.48}$	$53.89_{\pm 1.73}$	$57.36_{\pm 0.64}$	$\mathbf{36.11}_{\pm 1.22}$	$91.30_{\pm 0.96}$	$68.20_{\pm 0.62}$	$50.21_{\pm 0.39}$	57.93
IPD, S-P (1%)	$44.17_{\pm 2.32}$	$51.11_{\pm 2.14}$	$56.81_{\pm 2.37}$	$33.33_{\pm 0.66}$	$91.07_{\pm 0.69}$	$68.93_{\pm 0.63}$	$49.58_{\pm 1.36}$	56.43
IPD, S-P (0.1%)	$45.14_{\pm 0.87}$	$47.50_{\pm 0.72}$	$55.28_{\pm 1.73}$	$33.71_{\pm 1.37}$	$90.75_{\pm 0.26}$	$68.55_{\pm 0.80}$	$50.46_{\pm 0.84}$	55.91
IPD, S-P (0.01%)	$43.19_{\pm 2.37}$	$49.03_{\pm 1.05}$	$55.83_{\pm 1.82}$	$34.85_{\pm 0.76}$	$89.93_{\pm 0.86}$	$68.73_{\pm 0.74}$	$50.93_{\pm 0.95}$	56.07
IPD, T-P (1%)	$47.92_{\pm 1.10}$	$56.81_{\pm 1.68}$	$60.83_{\pm 1.50}$	$35.10_{\pm 1.53}$	$91.75_{\pm 0.39}$	$\mathbf{69.57}_{\pm 0.62}$	$51.35_{\pm 2.21}$	**59.05**
IPD, T-P (0.1%)	$46.81_{\pm 0.64}$	$54.58_{\pm 3.31}$	$\mathbf{61.39}_{\pm 2.55}$	$32.83_{\pm 1.16}$	$\mathbf{91.77}_{\pm 0.38}$	$67.74_{\pm 0.26}$	$\mathbf{51.98}_{\pm 1.20}$	58.16
IPD, T-P (0.01%)	$46.81_{\pm 2.06}$	$\mathbf{57.36}_{\pm 0.87}$	$60.97_{\pm 4.05}$	$32.95_{\pm 3.41}$	$91.38_{\pm 0.21}$	$69.54_{\pm 0.61}$	$46.72_{\pm 1.49}$	57.96
Qwen3-0.6B/8B on LIMO								
Vanilla	$6.25_{\pm 1.67}$	$15.00_{\pm 1.91}$	$10.42_{\pm 1.10}$	$8.96_{\pm 0.79}$	$71.07_{\pm 0.40}$	$37.82_{\pm 0.26}$	$22.35_{\pm 0.91}$	24.55
Direct Distillation	$8.19_{\pm 0.87}$	$16.53_{\pm 1.34}$	$10.28_{\pm 0.96}$	$8.21_{\pm 0.44}$	$71.25_{\pm 0.61}$	$38.55_{\pm 0.54}$	$17.38_{\pm 0.26}$	24.34
Student RFT	$4.31_{\pm 2.55}$	$15.14_{\pm 1.88}$	$9.58_{\pm 1.67}$	$8.71_{\pm 1.14}$	$69.07_{\pm 2.29}$	$37.50_{\pm 0.39}$	$24.12_{\pm 0.66}$	24.06
Teacher RFT	$7.78_{\pm 0.24}$	$16.39_{\pm 1.97}$	$16.11_{\pm 1.05}$	$11.87_{\pm 0.44}$	$74.43_{\pm 0.88}$	$41.23_{\pm 0.43}$	$24.96_{\pm 3.03}$	27.54
IPD, S-P (1%)	$8.33_{\pm 1.50}$	$18.75_{\pm 0.00}$	$14.58_{\pm 1.10}$	$\mathbf{14.65}_{\pm 0.58}$	$76.60_{\pm 0.74}$	$43.15_{\pm 0.31}$	$26.98_{\pm 1.71}$	29.01
IPD, S-P (0.1%)	$7.92_{\pm 0.42}$	$18.89_{\pm 0.48}$	$12.92_{\pm 0.72}$	$11.74_{\pm 0.76}$	$75.58_{\pm 0.29}$	$41.47_{\pm 0.32}$	$27.48_{\pm 1.90}$	28.00
IPD, S-P (0.01%)	$9.03_{\pm 0.64}$	$16.11_{\pm 1.05}$	$14.58_{\pm 1.50}$	$11.24_{\pm 1.87}$	$75.07_{\pm 0.42}$	$41.20_{\pm 0.49}$	$24.79_{\pm 0.98}$	27.43
IPD, T-P (1%)	$9.58_{\pm 1.91}$	$\mathbf{19.31}_{\pm 0.87}$	$14.31_{\pm 2.14}$	$12.75_{\pm 0.22}$	$\mathbf{77.98}_{\pm 0.40}$	$\mathbf{45.07}_{\pm 0.06}$	$28.70_{\pm 1.10}$	29.67
IPD, T-P (0.1%)	$9.31_{\pm 1.05}$	$17.92_{\pm 0.42}$	$\mathbf{16.25}_{\pm 1.10}$	$12.88_{\pm 1.00}$	$76.78_{\pm 0.65}$	$44.09_{\pm 0.60}$	$\mathbf{32.74}_{\pm 2.15}$	**30.00**
IPD, T-P (0.01%)	$\mathbf{11.53}_{\pm 1.46}$	$\mathbf{19.31}_{\pm 1.05}$	$15.00_{\pm 1.25}$	$13.51_{\pm 0.58}$	$77.22_{\pm 0.46}$	$43.79_{\pm 0.54}$	$28.03_{\pm 0.33}$	29.77
Qwen3-1.7B/8B on s1K								
Vanilla	$31.81_{\pm 2.29}$	$32.64_{\pm 1.68}$	$33.47_{\pm 2.44}$	$25.38_{\pm 1.37}$	$88.72_{\pm 0.28}$	$57.15_{\pm 1.00}$	$42.21_{\pm 1.07}$	44.48
Direct Distillation	$24.86_{\pm 0.96}$	$26.67_{\pm 0.72}$	$27.64_{\pm 2.51}$	$23.74_{\pm 1.58}$	$87.10_{\pm 0.10}$	$54.17_{\pm 0.13}$	$26.01_{\pm 1.46}$	38.60
Student RFT	$33.19_{\pm 1.88}$	$35.42_{\pm 2.20}$	$35.69_{\pm 1.58}$	$27.27_{\pm 1.00}$	$88.33_{\pm 0.45}$	$57.62_{\pm 0.41}$	$39.27_{\pm 0.91}$	45.26
Teacher RFT	$30.97_{\pm 1.73}$	$35.00_{\pm 0.42}$	$39.31_{\pm 2.06}$	$26.89_{\pm 1.74}$	$88.95_{\pm 0.18}$	$58.10_{\pm 0.39}$	$42.05_{\pm 0.55}$	45.90
IPD, S-P (1%)	$34.44_{\pm 0.87}$	$35.14_{\pm 0.48}$	$39.44_{\pm 0.64}$	$27.65_{\pm 2.37}$	$89.48_{\pm 0.13}$	$58.79_{\pm 0.79}$	$40.78_{\pm 1.34}$	46.53
IPD, S-P (0.1%)	$34.03_{\pm 2.96}$	$33.33_{\pm 1.67}$	$40.56_{\pm 2.41}$	$28.16_{\pm 0.44}$	$89.38_{\pm 0.51}$	$58.96_{\pm 0.39}$	$42.89_{\pm 0.81}$	46.76
IPD, S-P (0.01%)	$34.58_{\pm 1.25}$	$34.58_{\pm 1.25}$	$36.11_{\pm 2.93}$	$29.92_{\pm 0.66}$	$89.13_{\pm 0.38}$	$58.75_{\pm 0.32}$	$41.96_{\pm 2.08}$	46.44
IPD, T-P (1%)	$33.61_{\pm 1.97}$	$37.08_{\pm 1.10}$	$40.28_{\pm 1.05}$	$28.03_{\pm 1.65}$	$\mathbf{90.00}_{\pm 0.46}$	$59.57_{\pm 0.27}$	$\mathbf{47.05}_{\pm 0.72}$	47.95
IPD, T-P (0.1%)	$\mathbf{36.67}_{\pm 1.82}$	$\mathbf{37.50}_{\pm 2.32}$	$39.17_{\pm 2.92}$	$29.29_{\pm 2.58}$	$89.95_{\pm 0.46}$	$\mathbf{59.92}_{\pm 0.37}$	$44.32_{\pm 0.66}$	48.12
IPD, T-P (0.01%)	$34.44_{\pm 3.07}$	$37.08_{\pm 2.32}$	$\mathbf{41.81}_{\pm 1.88}$	$\mathbf{31.57}_{\pm 1.09}$	$89.78_{\pm 0.24}$	$59.80_{\pm 0.20}$	$45.29_{\pm 0.96}$	**48.54**

4.2 Threshold Robustness and Proposal Direction

The teacher-propose advantage does not depend on tuning the threshold. The threshold sets how often the student intervenes: in the primary setting, the strictest threshold makes the student intervene about five times as often as the most permissive one (Table 6). Yet T-P exceeds Teacher RFT in all twelve setting–threshold comparisons of Table 1, and its average changes little within a setting. Which threshold is best varies across training seeds, but the best average in every setting and seed comes from T-P (Appendix E.1). Even the most permissive threshold already replaces the proposals the student finds least likely, and intervening more often adds little. This is the pattern our hypothesis predicts: the obstacle lies in a few extreme tokens, not in the bulk of the teacher's trace.

Proposal direction, in contrast, matters. Student-propose IPD interleaves the same frozen models under the same thresholds and differs only in which model proposes. It also improves on Vanilla throughout, so interleaving helps in either direction, but even its best threshold falls below T-P's worst in every setting. When the student leads, the teacher contributes only corrections, and the traces remain mostly the student's own. S-P traces are shorter and solve fewer questions, whereas T-P traces are as long as the teacher's and solve nearly as many (Table 6). **At every tested threshold, the student learns more from adapted teacher reasoning than from corrected student reasoning.**

4.3 GAINS UNDER DATA AND BUDGET CONTROLS

IPD retains its advantage over additional sampling, RL, and OPD under comparable budgets. In the primary setting, increasing Student RFT from 8 to 100 attempts raises the number of correct full traces from 219 to 468, but average accuracy remains essentially unchanged (24.95 versus 24.88). IPD reaches 29.72 with 516 correct full traces, exceeding Teacher RFT at 27.65 despite its 572 traces (Tables 2 and 3). The quantity of verified solutions alone therefore does not explain its advantage.

Table 2: Sampling and training budgets for Qwen3-0.6B/8B on s1K. Relative budget weights the maximum rollouts by the sizes of the models that generate and score them, normalized to IPD (Appendix B.1). Every baseline except Student RFT-8 spends close to IPD's budget.

Method	Correct full traces	Training steps	Maximum rollouts	Relative budget
Student RFT-8	219	937	8,000	$0.07\times$
Student RFT-100	468	937	100,000	$0.87\times$
Teacher RFT	572	937	8,000	$0.93\times$
GRPO	N/A	1,000	128,000	$1.12\times$
OPD	N/A	500	8,000	$1.00\times$
IPD, T-P (0.1%)	516	937	8,000	$1.00\times$

Table 3: Additional-sampling and on-policy baselines for Qwen3-0.6B/8B on s1K, under the budgets of Table 2. Scoring follows Table 1; bold marks the best value per column. IPD has the best average, ahead of heavier sampling, GRPO, and OPD, and has the best value on five of the seven benchmarks.

Method	AIME24	AIME25	AIME26	HMMT26	MATH500	Olympiad	GPQA-D	Average
Vanilla	$6.25_{\pm 1.67}$	$15.00_{\pm 1.91}$	$10.42_{\pm 1.10}$	$8.96_{\pm 0.79}$	$71.07_{\pm 0.40}$	$37.82_{\pm 0.26}$	$22.35_{\pm 0.91}$	24.55
Direct Distillation	$4.86_{\pm 0.24}$	$18.33_{\pm 2.20}$	$10.14_{\pm 0.64}$	$8.46_{\pm 1.22}$	$71.63_{\pm 0.75}$	$35.87_{\pm 0.27}$	$19.78_{\pm 0.07}$	24.15
Student RFT-8	$6.11_{\pm 1.34}$	$14.44_{\pm 0.64}$	$10.28_{\pm 0.48}$	$7.70_{\pm 0.95}$	$71.42_{\pm 1.35}$	$38.24_{\pm 0.27}$	$26.43_{\pm 1.07}$	24.95
Student RFT-100	$3.89_{\pm 1.05}$	$16.39_{\pm 1.73}$	$10.83_{\pm 1.25}$	$9.34_{\pm 2.15}$	$71.45_{\pm 0.33}$	$38.70_{\pm 0.47}$	$23.53_{\pm 0.62}$	24.88
Teacher RFT	$7.50_{\pm 1.91}$	$19.86_{\pm 0.64}$	$11.11_{\pm 3.07}$	$11.87_{\pm 2.22}$	$74.40_{\pm 0.40}$	$41.31_{\pm 0.25}$	$27.48_{\pm 0.65}$	27.65
GRPO	$5.83_{\pm 2.20}$	$17.78_{\pm 1.92}$	$12.22_{\pm 1.68}$	$10.61_{\pm 1.37}$	$72.62_{\pm 0.35}$	$39.26_{\pm 0.67}$	$18.60_{\pm 0.84}$	25.27
OPD	$8.47_{\pm 1.68}$	$\mathbf{20.97}_{\pm 0.24}$	$\mathbf{15.28}_{\pm 1.46}$	$10.61_{\pm 1.52}$	$75.32_{\pm 0.28}$	$42.84_{\pm 0.26}$	$26.05_{\pm 1.42}$	28.51
IPD, T-P (0.1%)	$\mathbf{10.14}_{\pm 3.15}$	$20.56_{\pm 0.48}$	$14.03_{\pm 2.77}$	$\mathbf{12.37}_{\pm 1.09}$	$\mathbf{76.30}_{\pm 1.06}$	$\mathbf{42.89}_{\pm 0.85}$	$\mathbf{31.78}_{\pm 1.39}$	**29.72**

Another possible confound is the partial trace: for each question it fails to solve, IPD still trains on the first 128 tokens of an attempt. We test the effect of partial traces from both sides. First, we train IPD on its 516 full traces alone, and its average drops only slightly. Second, we give Direct Distillation the same treatment: we keep the original s1K traces either for the questions IPD solved or for the questions whose s1K trace our verifier accepts, and train each selection with and without 128-token partial traces for the remaining questions. Partial traces again add a little, yet every variant stays far below IPD (Appendix B.2, Tables 14 and 15). They help, but do not explain IPD's advantage.

For on-policy comparisons, we use GRPO (Shao et al., 2024) and OPD (Agarwal et al., 2024). GRPO learns from verified correctness rewards, while OPD minimizes reverse KL to the teacher on newly sampled student rollouts. Under comparable budgets, they average 25.27 and 28.51, both below IPD's 29.72; GRPO gains only 0.72 points over Vanilla. Neither correctness rewards nor dense teacher feedback on student rollouts matches the gains from IPD's interleaved targets. Appendix B.1 provides the baseline protocols and budget accounting. **IPD's gains are not explained by trace quantity or partial traces from failed attempts, and they exceed RL and OPD under comparable budgets.**

4.4 ABLATING TRAJECTORY CONSTRUCTION AND LEARNING TARGETS

During distillation, a teacher token the student would almost never produce causes two problems: the student learns to reproduce it, and later reasoning builds on it. IPD replaces only such tokens, lets the teacher continue, and trains on the result. Tables 4 and 5 test controls that drop these steps.

The first controls keep completed traces and change only their supervision. Trace filtering ranks s1K traces by mean surprisal, keeps the lower half as full traces, and cuts the rest to 128-token partial traces. Loss masking drops the loss on tokens whose student probability falls below 1%, 0.1%, or 0.01%. Neither regenerates anything: filtering discards the reasoning in the traces it cuts, and masking still trains on a continuation built on the masked token. Both fall short of IPD, so the reasoning after an unsupported token has to be revised, not only the token's supervision.

Table 4: Trace construction and learning target of each control. A token from the source passes the gate or is replaced by the fallback; q_β is the geometric mean of the two policies, CE is cross-entropy, and KL is forward KL divergence. Only IPD both gates generation and trains on the realized tokens.

Method	Token source	Selection or gate	Fallback	Supervision target
Trace filtering	Corpus	Trace surprisal	None	Corpus token (CE)
Loss masking	Corpus	$p_S(z) \geq \tau$ (loss only)	None	Unmasked token (CE)
Dist. Interpo.	q_β	None	None	Sampled token (CE)
SKD	Student	Teacher top-K	Teacher	Teacher dist. (KL)
IPD, S-P	Student	$p_T(z) \geq \tau$	Teacher	Realized token (CE)
IPD, T-P	Teacher	$p_S(z) \geq \tau$	Student	Realized token (CE)

The remaining controls generate new traces but change how the policies are combined or what the student learns. Distribution interpolation samples every token from $q_\beta \propto p_T^\beta p_S^{1-\beta}$ with $\beta = 0.5$, a geometric mean that blends the two policies everywhere but sets no floor on student probability. SKD also interleaves the policies, accepting student proposals among the teacher's top 25 tokens and replacing the rest with teacher samples, but it trains the student to match the teacher's distribution at every position (Xu et al., 2025b), so student tokens shape its prefixes without becoming targets. Both perform below T-P; interpolation, which weights the two policies equally, falls between the S-P and T-P averages of Table 1. Appendix B.3 gives their specifications.

Table 5: Results for the controls of Table 4, for Qwen3-0.6B/8B on s1K. Loss-masking percentages are probability thresholds. Scoring follows Table 1; bold marks the best value per column. No control reaches IPD's average: filtering and masking fall far behind, and interpolation and SKD come close.

Method	AIME24	AIME25	AIME26	HMMT26	MATH500	Olympiad	GPQA-D	Average
Vanilla	$6.25_{\pm 1.67}$	$15.00_{\pm 1.91}$	$10.42_{\pm 1.10}$	$8.96_{\pm 0.79}$	$71.07_{\pm 0.40}$	$37.82_{\pm 0.26}$	$22.35_{\pm 0.91}$	24.55
Trace filtering	$6.25_{\pm 1.10}$	$18.19_{\pm 0.48}$	$10.00_{\pm 0.72}$	$10.48_{\pm 2.69}$	$74.03_{\pm 1.04}$	$38.50_{\pm 1.34}$	$28.79_{\pm 1.08}$	26.61
Loss masking (1%)	$7.08_{\pm 0.42}$	$18.47_{\pm 1.05}$	$9.72_{\pm 1.05}$	$9.34_{\pm 1.79}$	$73.15_{\pm 0.18}$	$37.64_{\pm 0.47}$	$22.77_{\pm 2.44}$	25.45
Loss masking (0.1%)	$6.11_{\pm 0.24}$	$17.50_{\pm 1.10}$	$10.28_{\pm 2.44}$	$10.35_{\pm 3.09}$	$72.55_{\pm 0.25}$	$36.82_{\pm 0.23}$	$22.64_{\pm 1.68}$	25.18
Loss masking (0.01%)	$6.67_{\pm 1.25}$	$18.33_{\pm 1.10}$	$9.58_{\pm 0.00}$	$7.83_{\pm 1.16}$	$71.90_{\pm 0.95}$	$36.86_{\pm 0.25}$	$23.06_{\pm 0.96}$	24.89
Dist. Interpo.	$10.00_{\pm 0.72}$	$19.03_{\pm 1.05}$	$14.72_{\pm 1.58}$	$\mathbf{13.26}_{\pm 1.97}$	$\mathbf{77.82}_{\pm 0.68}$	$\mathbf{43.71}_{\pm 0.09}$	$28.03_{\pm 1.20}$	29.51
SKD	$\mathbf{13.19}_{\pm 1.20}$	$19.58_{\pm 1.25}$	$\mathbf{15.83}_{\pm 0.83}$	$12.50_{\pm 1.00}$	$69.08_{\pm 0.98}$	$40.69_{\pm 0.65}$	$\mathbf{32.95}_{\pm 0.51}$	29.12
IPD, T-P (0.1%)	$10.14_{\pm 3.15}$	$\mathbf{20.56}_{\pm 0.48}$	$14.03_{\pm 2.77}$	$12.37_{\pm 1.09}$	$76.30_{\pm 1.06}$	$42.89_{\pm 0.85}$	$31.78_{\pm 1.39}$	**29.72**

Controls that drop any of these choices fall short: regenerating the continuation, replacing selectively, and learning from the generated tokens each contribute. **The strongest transfer comes from revising teacher reasoning during generation and retaining both policies' contributions as SFT targets.**

5 What IPD Changes and What It Preserves

5.1 IPD Reduces Extreme-Tail Mismatch

Teacher-propose IPD keeps the teacher's reasoning while removing its rare, severe mismatches with the student. Figure 3 follows each dataset's upper-tail mean surprisal as the tail narrows, beside the gain of the student trained on it; Table 6 gives the diagnostics for the primary setting. The curves fan out toward the extreme tail, where Direct Distillation is highest, Teacher RFT next, and T-P lower in every setting. Student RFT is lower still, since its traces are the student's own.

Table 6: Dataset diagnostics for Qwen3-0.6B/8B on s1K. # Correct counts verified full traces; median length covers them, or all source traces for Direct Distillation. T-P keeps most of Teacher RFT's correct traces while lowering the mean of the top 0.01% token surprisals.

Condition	# Correct	Med. len. (tok)	Fallback (%)	Mean surp. (nat)	Top 0.01% surp. (nat)	Eval. avg.
Direct Distillation	479	9,549.5	N/A	1.1229	21.9850	24.15
Student RFT	219	6,133.0	N/A	0.2366	3.7291	24.95
Teacher RFT	572	9,211.0	N/A	0.4400	13.1366	27.65
IPD, S-P (1%)	455	7,537.0	1.795	0.3584	11.5742	29.10
IPD, S-P (0.1%)	367	7,026.0	0.969	0.3616	10.5838	28.34
IPD, S-P (0.01%)	312	7,488.5	0.542	0.3732	9.9513	28.73
IPD, T-P (1%)	503	9,391.0	0.728	0.4182	5.4950	29.58
IPD, T-P (0.1%)	516	9,583.5	0.241	0.4715	7.0902	29.72
IPD, T-P (0.01%)	532	9,371.0	0.136	0.4937	9.0937	29.91

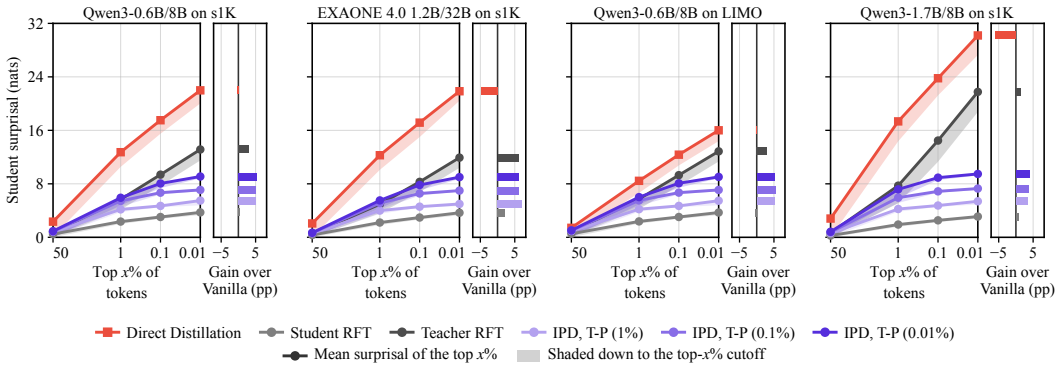


Figure 3: Student-surprisal tails and gains in four settings. Curves show the mean of the top $x\%$ token surprisals, shaded to the top-$x\%$ cutoff; bars show gains over Vanilla at the mean of the top 0.01%. Curves separate as x narrows; T-P, below Direct Distillation and Teacher RFT, gains most.

Transfer follows the same order in all four settings: Direct Distillation leaves the student below Vanilla, Teacher RFT improves it, and T-P improves it most. Student RFT has the lowest extreme-tail mean surprisal, yet its self-generated traces train weaker students than either IPD direction in every setting. T-P keeps the teacher's learning signal, as the teacher writes almost every token, yet its extreme-tail mean surprisal falls well below Teacher RFT's. Neither correct-trace count, median length, nor mean surprisal explains this: the best configuration has neither the most correct traces nor the longest ones, and S-P datasets have lower mean surprisal than T-P datasets yet train weaker students (Tables 6 and 16). **Low extreme-tail surprisal supports transfer when paired with an external learning signal.**

5.2 Teacher Proposals Add Value Beyond Compatibility

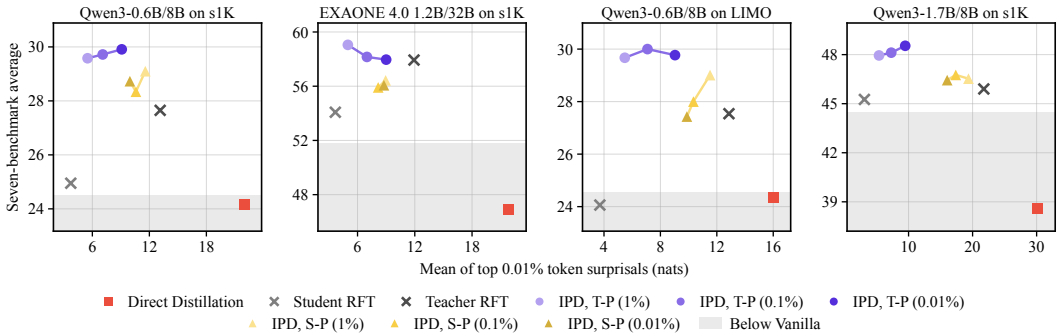


Figure 4: Student performance against the mean of the top 0.01% token surprisals of its training set. Each point is one training set; lines connect the thresholds of each IPD direction, and shading marks averages below Vanilla. T-P sits above S-P in every setting, even where the two overlap in surprisal.

The external learning signal also separates the two IPD directions. T-P outperforms S-P in every setting, even where their extreme-tail mean surprisals overlap (Figure 4). EXAONE shows this directly: one T-P dataset has higher extreme-tail mean surprisal than every S-P dataset, yet trains a better student than any of them.

In T-P, the teacher writes almost every token; in S-P, it writes only the rare fallbacks (Table 6). S-P data thus carry the external signal only as corrections to the student's own reasoning, whereas T-P data carry the teacher's reasoning throughout. **More teacher supervision strengthens transfer at comparable extreme-tail surprisal.**

5.3 Token Replacements and Reasoning Quality

Student fallbacks steer the teacher toward plainer continuations, and the resulting traces, judged slightly weaker than s1K as training data, train a student that reasons better. Table 7 classifies each

rejected teacher proposal and its student fallback, from surface form to the discourse pivots that shift reasoning stages (Appendix C.2). Fallbacks shift mass toward surface form and discourse connectives, and away from reasoning-content words and pivots. Where the teacher proposes a pivot such as "Wait", the student often continues the argument with a connective such as "Similarly" or "Hence".

Table 7: Token-category shares, T-P ($\tau = 0.1\%$). Subscripts: relative change from rejected tokens.

Category	Rejected T. tokens (%)	Fallback S. tokens (%)
Surface form	24.59	$34.16_{+39\%}$
Computational content	14.16	$15.81_{+12\%}$
Reasoning content	57.30	$44.45_{-22\%}$
Discourse connective	2.82	$5.26_{+87\%}$
Discourse pivot	1.13	$0.32_{-72\%}$

Table 8: IPD win rate (%) against s1K, mean $\pm$ sample standard deviation across three judges.

Criterion	**Training traces** IPD data vs. s1K	**Sampled traces** IPD- vs. s1K-trained
Problem groundedness	$54.4_{\pm 2.3}$	$55.8_{\pm 2.2}$
Step coherence	$51.9_{\pm 2.9}$	$56.2_{\pm 2.1}$
Logical validity	$50.3_{\pm 2.7}$	$54.4_{\pm 2.9}$
No redundancy	$41.1_{\pm 2.6}$	$50.1_{\pm 1.9}$
Overall preference	$47.7_{\pm 2.9}$	$54.1_{\pm 2.3}$

Table 8 reports blind judgments on both-correct pairs: IPD against s1K training data, and traces sampled from the students trained on each (Appendix C.3). As training data, IPD traces are better grounded but less concise than s1K, and s1K is slightly preferred overall. The IPD-trained student is then preferred overall and on groundedness, coherence, and validity, and is as concise. **IPD trades a little trace quality for compatibility and trains a student that reasons better.**

5.4 IPD Broadens Problem-Solving Coverage

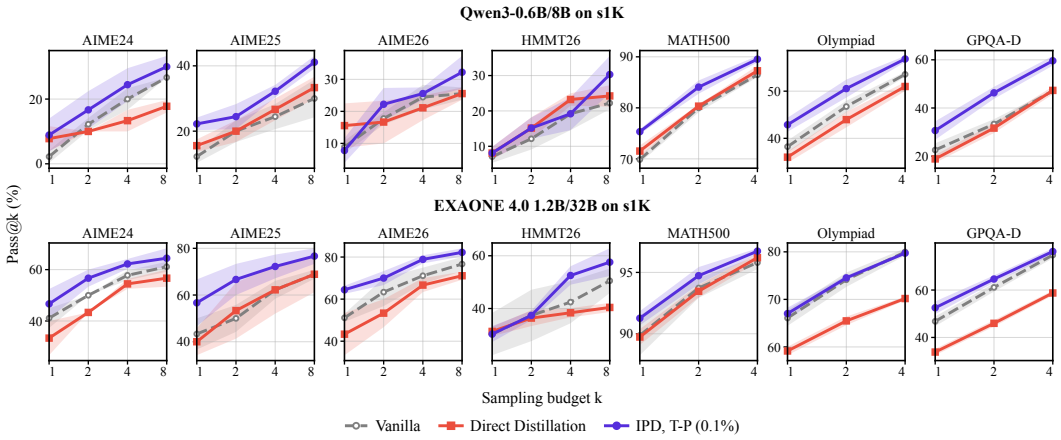


Figure 5: Pass@k under matched decoding on two s1K settings. Curves average three seeds; bands show one sample standard deviation. IPD leads both baselines at the largest budget in 13 of 14 panels.

IPD expands the range of problems the student can solve on its own. Pass@k measures whether any of k sampled solutions is correct, which is what limits RL: it learns from a problem only when the student produces a correct rollout. At the largest matched sampling budget, IPD leads Vanilla and Direct Distillation in 27 of 28 benchmark–setting pairs (Figure 5; Tables 23–26). IPD therefore gives RL more successes to reinforce and OPD more useful prefixes for teacher feedback. **Broader problem-solving coverage motivates IPD as an initialization for subsequent on-policy learning.**

6 Conclusion

IPD builds SFT targets in which the student's own tokens replace teacher proposals it would rarely produce, and the teacher continues from the revised prefix. Across model families, datasets, and thresholds, IPD outperforms direct distillation and RFT. Transfer failures trace to the few tokens the student finds least likely, while teacher-supplied tokens add value beyond compatibility. Broader problem-solving coverage suggests IPD could initialize subsequent RL and OPD. **Small models learn reasoning paths better when those paths arrive in a form they can make their own.**

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A Experimental Setup

A.1 Training-Data Construction

Table 9 summarizes the generation controls for the 1,000-row s1K and 800-row LIMO sources. Student RFT, Teacher RFT, and IPD use an eight-attempt budget per question. The correctness verifier is an LLM judge, `gpt-5.4-nano-2026-03-17`. Each unsolved IPD question contributes a partial trace of at most 128 tokens from its latest failed attempt.

Direct Distillation preserves all released source traces. Our verifier accepts 479 of the 1,000 s1K-1.1 traces as correct; the s1 paper reports 630 (63.0%) (Muennighoff et al., 2025). LIMO-v2 provides reasoning-only traces, with a separate answer used for judging.

Table 9: Dataset-generation controls for verified full traces and partial traces. Each IPD dataset covers all source questions with targets generated by the same IPD condition.

Setting	**Value**
Prompt-token budget	4,096
Generated-token budget	16,384
Temperature / top-p	0.6 / 0.95
Full-trace attempts	At most 8
Partial-trace length	At most 128 tokens
Unsolved row	Condition-native partial trace
Generation close	No forced reasoning close
Source coverage	1,000 rows for s1K; 800 rows for LIMO

A.2 Supervised Fine-Tuning

Table 10 lists the shared SFT controls. All headline comparisons use full-parameter BF16 SFT with 20,000-token packing blocks, prompt-loss masking, a microbatch of one, gradient accumulation of 16, and AdamW with a cosine schedule. Verified generated full traces receive the student's native end marker, whereas partial traces end abruptly. LIMO Direct Distillation is the reasoning-only exception described above. Within a headline setting, every method uses the same selected schedule.

Table 10: SFT controls for Qwen3 and EXAONE. All SFT methods within a setting share the same optimization schedule and prompt loss masking.

Setting	**Value**
Training type / precision	Full-parameter SFT / BF16
Packing block	20,000 tokens
Microbatch / accumulation / effective batch	1 / 16 / 16
Optimizer / schedule	AdamW / cosine
Loss	Prompt-masked next-token likelihood
Qwen3 headline schedule	15 epochs, 3×10^{-5}
EXAONE headline schedule	30 epochs, 1×10^{-5}
Complete trace end	Native EOS or turn-end marker
LIMO Direct Distillation end	Reasoning close, then no answer or EOS
Partial-trace end	Abrupt end with no marker

A.3 Held-Out Hyperparameter Search

We conduct a separate SFT hyperparameter search for each of the four headline settings. In each search, we reserve a deterministic 10% of the dataset and train on the remaining 90%. The search uses Teacher RFT as the representative off-policy condition and T-P ($\tau = 0.1\%$) as the representative interleaved-policy condition. These are the highest-performing conditions in their respective families across multiple headline settings. We therefore use them to select one shared schedule per model–dataset setting rather than tuning every method separately.

The grid crosses 5, 15, and 30 epochs with learning rates 1×10^{-5}, 2×10^{-5}, and 3×10^{-5}. We select the schedule that ranks near the top for both representative conditions on both held-out Avg@8

and held-out Pass@8, rather than the argmax of either condition alone. In the two grids shown, the representative conditions favor the same region of the grid, the higher learning rates for Qwen3 and the lower ones for EXAONE, so the shared schedule sits near the optimum of each rather than compromising between conflicting optima. For Qwen3-0.6B/8B the selected cell is second on Avg@8 for both representative conditions, and on Pass@8 it is second for IPD and joint third for Teacher RFT; for EXAONE 4.0 1.2B/32B it is first on Avg@8 and second on Pass@8 for both.

Tables 11 and 12 present representative search grids for Qwen3-0.6B/8B on s1K and EXAONE on s1K. The Qwen3-0.6B/LIMO and Qwen3-1.7B/s1K searches independently select the same 15-epoch, 3×10^{-5} schedule as Qwen3-0.6B/s1K. Table 13 summarizes the selected schedules for all four settings.

Table 11: Held-out schedule search for Qwen3-0.6B/8B on s1K using a deterministic 90/10 split. Shading marks the schedule selected for all SFT methods within this setting.

Method	**LR**	**Epochs**	**Avg@8**	**Pass@1**	**Pass@2**	**Pass@4**	**Pass@8**
Teacher RFT	1×10^{-5}	5	11.00	10.00	16.00	25.00	32.00
Teacher RFT	1×10^{-5}	15	12.88	10.00	17.00	26.00	37.00
Teacher RFT	1×10^{-5}	30	11.62	8.00	19.00	27.00	37.00
Teacher RFT	2×10^{-5}	5	11.75	14.00	23.00	25.00	39.00
Teacher RFT	2×10^{-5}	15	13.12	12.00	18.00	24.00	34.00
Teacher RFT	2×10^{-5}	30	15.00	14.00	23.00	38.00	40.00
Teacher RFT	3×10^{-5}	5	13.88	15.00	22.00	30.00	41.00
Teacher RFT	3×10^{-5}	15	15.25	11.00	23.00	32.00	39.00
Teacher RFT	3×10^{-5}	30	15.88	15.00	23.00	34.00	38.00
IPD, T-P (0.1%)	1×10^{-5}	5	14.50	13.00	19.00	30.00	39.00
IPD, T-P (0.1%)	1×10^{-5}	15	13.63	12.00	21.00	32.00	42.00
IPD, T-P (0.1%)	1×10^{-5}	30	15.00	16.00	22.00	35.00	41.00
IPD, T-P (0.1%)	2×10^{-5}	5	15.25	13.00	24.00	32.00	40.00
IPD, T-P (0.1%)	2×10^{-5}	15	14.50	15.00	25.00	33.00	41.00
IPD, T-P (0.1%)	2×10^{-5}	30	17.88	13.00	25.00	35.00	47.00
IPD, T-P (0.1%)	3×10^{-5}	5	15.62	14.00	21.00	30.00	41.00
IPD, T-P (0.1%)	3×10^{-5}	15	17.50	21.00	32.00	39.00	46.00
IPD, T-P (0.1%)	3×10^{-5}	30	14.75	17.00	24.00	32.00	42.00

Table 12: Held-out schedule search for EXAONE 4.0 1.2B/32B on s1K using a deterministic 90/10 split. Shading marks the schedule selected for all SFT methods within this setting.

Method	**LR**	**Epochs**	**Avg@8**	**Pass@1**	**Pass@2**	**Pass@4**	**Pass@8**
Teacher RFT	1×10^{-5}	5	43.12	40.00	55.00	60.00	66.00
Teacher RFT	1×10^{-5}	15	43.00	43.00	53.00	61.00	61.00
Teacher RFT	1×10^{-5}	30	43.62	47.00	54.00	59.00	66.00
Teacher RFT	2×10^{-5}	5	41.62	40.00	52.00	63.00	67.00
Teacher RFT	2×10^{-5}	15	41.00	41.00	54.00	59.00	65.00
Teacher RFT	2×10^{-5}	30	40.38	45.00	50.00	56.00	61.00
Teacher RFT	3×10^{-5}	5	42.50	48.00	54.00	61.00	65.00
Teacher RFT	3×10^{-5}	15	41.00	38.00	46.00	59.00	61.00
Teacher RFT	3×10^{-5}	30	37.88	40.00	52.00	57.00	61.00
IPD, T-P (0.1%)	1×10^{-5}	5	41.12	43.00	52.00	60.00	63.00
IPD, T-P (0.1%)	1×10^{-5}	15	41.88	46.00	53.00	57.00	61.00
IPD, T-P (0.1%)	1×10^{-5}	30	44.75	46.00	55.00	60.00	65.00
IPD, T-P (0.1%)	2×10^{-5}	5	44.25	49.00	54.00	62.00	67.00
IPD, T-P (0.1%)	2×10^{-5}	15	43.50	45.00	53.00	59.00	63.00
IPD, T-P (0.1%)	2×10^{-5}	30	42.88	37.00	50.00	60.00	63.00
IPD, T-P (0.1%)	3×10^{-5}	5	42.00	41.00	52.00	58.00	64.00
IPD, T-P (0.1%)	3×10^{-5}	15	39.88	41.00	50.00	55.00	60.00
IPD, T-P (0.1%)	3×10^{-5}	30	39.25	36.00	46.00	55.00	59.00

Table 13: SFT schedules selected by independent held-out searches in all four settings. Within each setting, all SFT methods use the selected schedule.

Headline setting	**Epochs**	**Learning rate**
Qwen3-0.6B/8B on s1K	15	3×10^{-5}
Qwen3-0.6B/8B on LIMO	15	3×10^{-5}
Qwen3-1.7B/8B on s1K	15	3×10^{-5}
EXAONE 4.0 1.2B/32B on s1K	30	1×10^{-5}

A.4 Evaluation and Statistical Reporting

Evaluation permits 32,768 generated tokens, reserving the last 2,048 tokens for the answer after at most 30,720 reasoning tokens. Sampling uses temperature 0.6 and top-p 0.95. Attempt k uses the evaluation base seed plus k.

For each checkpoint, Avg@k averages accuracy over k sampled attempts. For each fine-tuned condition, seeds 42, 1, and 2026 index three independent SFT training runs. Table 1 reports the mean and sample standard deviation of their benchmark scores. These training seeds are distinct from evaluation sampling seeds. Pass@k reports cumulative question-level coverage at the available sampling budgets without a per-seed standard deviation.

B Baselines and Ablation Controls

B.1 On-Policy Baselines and Budget Accounting

Both baselines initialize from the released Qwen3-0.6B checkpoint and train on the s1K questions with a learning rate of 10^{-6}. Generation uses a 4,096-token prompt allowance, a 16,384-token output limit, temperature 0.6, and top-p 0.95. Each update uses 16 questions, with fresh rollouts sampled from the current student.

GRPO (Shao et al., 2024) samples eight rollouts per question and uses binary answer-correctness rewards to compute group-relative advantages. It performs one optimization pass per rollout batch and uses a KL penalty of 0.01 against the frozen initial student.

OPD (Agarwal et al., 2024) samples one rollout per question and minimizes token-level reverse KL against the frozen Qwen3-8B teacher. The objective uses the teacher's top-20 tokens with a stop-gradient normalizer. All sampled rollouts contribute to the loss regardless of correctness, including incorrect full rollouts.

In the primary setting, RFT and IPD use 15 SFT epochs on 1,000 questions with an effective batch size of 16. This gives $1{,}000 \times 15/16 = 937.5$ batches; dropping the final half-batch yields the 937 optimizer updates reported in Table 2.

IPD permits eight attempts per question, giving at most $8 \times 1{,}000 = 8{,}000$ rollouts. GRPO generates eight rollouts per question for 16 questions at each of 1,000 steps, totaling $8 \times 16 \times 1{,}000 = 128{,}000$. OPD generates one rollout per question for 16 questions at each of 500 steps, totaling $16 \times 500 = 8{,}000$. We weight maximum rollout counts by the sizes of the models used for generation and teacher scoring. The parameter-weighted rollout budgets are therefore $(0.6{+}8) \times 8{,}000 = 68{,}800$ for IPD and OPD and $0.6 \times 128{,}000 = 76{,}800$ for GRPO, in billions of parameters times rollouts. Normalizing by IPD gives $1.00\times$ for OPD and $1.12\times$ for GRPO. The same weighting gives $0.6 \times 8{,}000 = 4{,}800$ for Student RFT-8, $0.6 \times 100{,}000 = 60{,}000$ for Student RFT-100, and $8 \times 8{,}000 = 64{,}000$ for Teacher RFT, or $0.07\times$, $0.87\times$, and $0.93\times$.

B.2 Question-Matched and Correctness-Based Controls

Question-matched controls retain original s1K traces for the 516 questions solved by T-P at $\tau = 0.1\%$. The full-only variant uses these 516 traces alone; the full-plus-partial variant adds 128-token partial traces for the remaining 484 questions. Correctness-based controls instead retain the 479 source traces accepted by our verifier, either alone or with 128-token partial traces from the 521 rejected traces. All Direct Distillation partial traces come from the corresponding original s1K traces.

We additionally evaluate IPD using only its 516 verified full traces. Comparing full-only and full-plus-partial variants measures the contribution of partial traces, while the question-matched comparison holds the selected questions and full-trace counts fixed. In both controls, adding partial traces does not close the gap to IPD, and IPD trained on its full traces alone reaches 29.63 against 29.72 with partial traces, so partial traces from failed attempts are not the main driver of IPD's gains. Tables 14 and 15 report dataset composition and downstream performance.

Table 14: Direct Distillation controls on s1K. Q-match selects questions solved by IPD, T-P (0.1%); C-based selects verifier-accepted source traces. Partial traces come from the remaining source traces.

Control	Full traces	128-token partial traces	Total rows
Q-match Direct, full-only	516	0	516
Q-match Direct	516	484	1,000
C-based Direct, full-only	479	0	479
C-based Direct	479	521	1,000

Table 15: Benchmark accuracy of the Direct Distillation controls in Table 14 and IPD with and without partial traces, for Qwen3-0.6B/8B on s1K. Scoring and bold marking follow Table 1.

Method	AIME24	AIME25	AIME26	HMMT26	MATH500	Olympiad	GPQA-D	Average
Vanilla	$6.25_{\pm 1.67}$	$15.00_{\pm 1.91}$	$10.42_{\pm 1.10}$	$8.96_{\pm 0.79}$	$71.07_{\pm 0.40}$	$37.82_{\pm 0.26}$	$22.35_{\pm 0.91}$	24.55
Direct Distillation	$4.86_{\pm 0.24}$	$18.33_{\pm 2.20}$	$10.14_{\pm 0.64}$	$8.46_{\pm 1.22}$	$71.63_{\pm 0.75}$	$35.87_{\pm 0.27}$	$19.78_{\pm 0.07}$	24.15
Q-match Direct, full-only	$5.28_{\pm 0.64}$	$16.94_{\pm 1.34}$	$8.89_{\pm 0.24}$	$9.85_{\pm 0.66}$	$72.72_{\pm 1.17}$	$38.06_{\pm 0.38}$	$21.34_{\pm 0.77}$	24.72
Q-match Direct	$5.69_{\pm 2.96}$	$17.36_{\pm 1.27}$	$8.89_{\pm 1.97}$	$10.73_{\pm 1.87}$	$73.22_{\pm 0.16}$	$37.33_{\pm 0.44}$	$23.02_{\pm 1.17}$	25.18
C-based Direct, full-only	$5.56_{\pm 0.48}$	$16.81_{\pm 0.87}$	$10.97_{\pm 1.68}$	$10.86_{\pm 1.22}$	$73.20_{\pm 1.30}$	$38.37_{\pm 0.37}$	$22.43_{\pm 0.64}$	25.46
C-based Direct	$7.50_{\pm 1.25}$	$17.08_{\pm 1.82}$	$9.58_{\pm 0.83}$	$10.48_{\pm 0.44}$	$72.15_{\pm 0.48}$	$38.24_{\pm 0.33}$	$26.14_{\pm 1.22}$	25.88
Student RFT-8	$6.11_{\pm 1.34}$	$14.44_{\pm 0.64}$	$10.28_{\pm 0.48}$	$7.70_{\pm 0.95}$	$71.42_{\pm 1.35}$	$38.24_{\pm 0.27}$	$26.43_{\pm 1.07}$	24.95
Student RFT-100	$3.89_{\pm 1.05}$	$16.39_{\pm 1.73}$	$10.83_{\pm 1.25}$	$9.34_{\pm 2.15}$	$71.45_{\pm 0.33}$	$38.70_{\pm 0.47}$	$23.53_{\pm 0.62}$	24.88
Teacher RFT	$7.50_{\pm 1.91}$	$19.86_{\pm 0.64}$	$11.11_{\pm 3.07}$	$11.87_{\pm 2.22}$	$74.40_{\pm 0.40}$	$41.31_{\pm 0.25}$	$27.48_{\pm 0.65}$	27.65
IPD, T-P (0.1%), full-only	$\mathbf{11.67}_{\pm 1.44}$	$18.61_{\pm 1.58}$	$13.75_{\pm 1.91}$	$11.99_{\pm 2.28}$	$76.25_{\pm 0.79}$	$42.67_{\pm 0.15}$	$\mathbf{32.49}_{\pm 1.29}$	29.63
IPD, T-P (0.1%)	$10.14_{\pm 3.15}$	$\mathbf{20.56}_{\pm 0.48}$	$\mathbf{14.03}_{\pm 2.77}$	$\mathbf{12.37}_{\pm 1.09}$	$\mathbf{76.30}_{\pm 1.06}$	$\mathbf{42.89}_{\pm 0.85}$	$31.78_{\pm 1.39}$	**29.72**

B.3 Trajectory and Learning-Target Ablations

Table 4 organizes the controls of Section 4.4. Trace filtering and loss masking keep completed traces and change only their supervision; distribution interpolation and SKD generate new traces under a different sampling rule or learning target.

Trace filtering ranks all 1,000 s1K traces by mean target-token surprisal under the frozen Qwen3-0.6B student, breaking ties by source index. The lower-surprisal 500 retain their full traces and termination markers; the remaining 500 retain at most 128 generated tokens and end abruptly without EOS. This condition selects by compatibility, not by answer correctness. Loss masking instead retains all full traces and masks a target whenever its frozen-student probability falls below the specified threshold. Structural and termination tokens are subject to the same rule, while masked tokens remain in the context used to predict later targets.

Distribution interpolation uses the IPD full-generation manifest and condition-native partial traces for unsolved questions, with the same attempt budget, verifier, and SFT schedule. Its candidate set is the union of the teacher's and student's top-20 tokens. Each token receives weight $p_T^{\beta} p_S^{1-\beta}$ with $\beta = 0.5$, normalized over that set, before applying the shared sampling controls. This symmetric setting weights the teacher and student equally. Candidate truncation can exclude tokens, but it does not impose IPD's explicit student-probability floor. The frozen references and common pipeline make the sampling rule the primary intervention.

SKD is a broader alternative rather than a single-component IPD ablation: it changes proposal direction, uses a teacher top-K acceptance rule with $K = 25$, and applies a forward-KL distillation target at every position, including those with student-authored tokens. IPD instead trains on each realized token with cross-entropy, so student tokens become supervision targets in both directions. The SKD comparison therefore concerns the overall design.

C Student Compatibility and Reasoning Quality

C.1 Dataset Diagnostics and Student Surprisal

Table 16 extends Table 6 to the other three settings. Fallback rate and mean surprisal average row-level values over all rows, whereas upper-tail means pool the target tokens, excluding prompts and padding, so a dataset's row-averaged mean surprisal can differ from its token-pooled mean.

Table 16: Dataset diagnostics for the other three settings. Columns follow Table 6, and surprisal is scored with each setting's frozen initial student.

Condition	# Correct	Med. len. (tok)	Fallback (%)	Mean surp. (nat)	Top 0.01% surp. (nat)	Eval. avg.
EXAONE 4.0 1.2B/32B on s1K						
Direct Distillation	479	10,148.0	N/A	1.0045	21.8501	46.92
Student RFT	531	9,461.0	N/A	0.1813	3.6722	54.08
Teacher RFT	554	11,494.0	N/A	0.3300	11.9172	57.93
IPD, S-P (1%)	541	9,787.0	0.535	0.2487	8.9435	56.43
IPD, S-P (0.1%)	520	9,863.0	0.182	0.2715	8.1635	55.91
IPD, S-P (0.01%)	498	9,830.0	0.087	0.2844	8.7646	56.07
IPD, T-P (1%)	529	11,430.0	0.272	0.3248	4.9858	59.05
IPD, T-P (0.1%)	523	11,464.0	0.077	0.3669	6.9855	58.16
IPD, T-P (0.01%)	512	11,637.5	0.043	0.3802	9.0012	57.96
Qwen3-0.6B/8B on LIMO						
Direct Distillation	737	12,093.5	N/A	0.7569	16.0060	24.34
Student RFT	203	4,448.0	N/A	0.2351	3.7068	24.06
Teacher RFT	404	11,025.5	N/A	0.4260	12.8592	27.54
IPD, S-P (1%)	270	7,457.5	1.498	0.3403	11.5197	29.01
IPD, S-P (0.1%)	206	5,043.0	0.806	0.3383	10.3399	28.00
IPD, S-P (0.01%)	193	5,058.0	0.440	0.3527	9.8774	27.43
IPD, T-P (1%)	309	10,777.0	0.486	0.4013	5.4746	29.67
IPD, T-P (0.1%)	305	10,903.0	0.153	0.4477	7.0838	30.00
IPD, T-P (0.01%)	312	10,371.5	0.083	0.4688	9.0348	29.77
Qwen3-1.7B/8B on s1K						
Direct Distillation	479	9,549.5	N/A	1.3475	30.1942	38.60
Student RFT	417	7,944.0	N/A	0.1114	3.1027	45.26
Teacher RFT	572	9,211.0	N/A	0.4313	21.7522	45.90
IPD, S-P (1%)	473	7,761.0	1.291	0.2381	19.3579	46.53
IPD, S-P (0.1%)	440	7,642.0	0.681	0.2134	17.3729	46.76
IPD, S-P (0.01%)	435	7,576.0	0.351	0.1998	16.0313	46.44
IPD, T-P (1%)	504	9,391.5	1.265	0.2944	5.3860	47.95
IPD, T-P (0.1%)	519	9,451.0	0.645	0.3851	7.2953	48.12
IPD, T-P (0.01%)	526	9,431.0	0.347	0.4487	9.4782	48.54

C.2 Token-Replacement Analysis

Table 7 classifies every fallback position in the 516 complete T-P ($\tau = 0.1\%$) training traces, pairing each rejected teacher proposal with the committed student fallback; partial traces are excluded because a 128-token fragment does not establish a reasoning stage. Classification applies to whole words: subword pieces inherit their word's class, and a rejected teacher token that begins a word is completed by sampling the teacher's continuation. Five classes are assigned by a fixed precedence, so they are exclusive and exhaustive: *surface form* (markup, delimiters, punctuation, whitespace, enumeration labels, and closed-class function words), *computational content* (numerals, identifiers, operators, and mathematical commands inside a derivation), *reasoning content* (the remaining open-class content words), *discourse connective* (a word linking steps within a reasoning stage), and *discourse pivot* (a word shifting the stage, of type realization, verification, exploration, or integration). Pivot types and the stage-shift criterion follow Lippmann & Yang (2025); connectives follow the Penn Discourse Treebank (Prasad et al., 2008).

Fixed rules label surface form and computational content directly, and flag words from a fixed discourse lexicon as candidate connectives or pivots. Only these candidates reach an LLM judge, `gpt-5.6-luna`, which sees the prefix, the candidate word, and its continuation, blind to its source, and decides whether the word links clauses in context, whether it is a pivot or a connective, and the pivot type. Of the 39 rejected teacher proposals labeled as pivots, 18 are replaced by student connectives.

C.3 TRACE-QUALITY EVALUATION

Table 8 measures reasoning quality with correctness held fixed. Training traces are paired on the 387 s1K questions for which both IPD (T-P, $\tau = 0.1\%$) and s1K supply a verified-correct full solution. Sampled traces pair the student trained on that IPD data with the Direct Distillation student trained on s1K, both at the headline schedule with training seed 42, on benchmark questions that each student answers correctly in at least one natively terminated attempt at evaluation seed 42; one such attempt per student is drawn uniformly. Every eligible AIME, HMMT, and GPQA-D question enters, and 80 questions each are drawn from MATH500 and OlympiadBench so that neither dominates. Judges see the complete reasoning and final answer.

Three LLM judges, `gpt-6-sol`, `claude-opus-5.5`, and `gemini-3.8-flash`, return a binary verdict, with no tie option, on four criteria and an overall preference, following the taxonomy of Lee & Hockenmaier (2025): *problem groundedness*, whether the trace uses only the quantities and conditions given in the problem and answers the question asked, following the hallucination score of ROSCOE (Golovneva et al., 2023) and the relevance label of REVEAL (Jacovi et al., 2024); *step coherence*, whether every step's premises were established earlier in the trace or given in the problem, following the missing-step error of ROSCOE; *logical validity*, whether every step the final argument relies on is correct in calculation and inference, following step-level correctness labels (Lightman et al., 2024; Xia et al., 2025); and *no redundancy*, whether every step advances the solution, following the neutral-step label of ReasonEval and the redundancy and repetition scores of ROSCOE. The rubric keeps the criteria exclusive: a false claim counts against validity, an unjustified true claim against coherence, and a retracted attempt against redundancy rather than validity.

Each pair is presented once with anonymized method labels, and the A/B order is randomized, balanced within the training set and within each benchmark (Zheng et al., 2023). The win rate is the share of pairs whose verdict favors IPD. Table 8 reports the mean and sample standard deviation of the three judges' win rates over the pairs all three scored, and Table 17 reports each judge.

Table 17: IPD win rate (%) against s1K by judge, for training traces and for traces sampled from the trained students. Table 8 reports the mean across judges.

Criterion	**Training traces** IPD data vs. s1K	**Sampled traces** IPD- vs. s1K-trained
`gpt-6-sol`		
Problem groundedness	55.4	56.2
Step coherence	50.8	53.8
Logical validity	47.7	51.0
No redundancy	38.1	48.6
Overall preference	45.9	51.4
`claude-opus-5.5`		
Problem groundedness	56.0	57.8
Step coherence	55.2	57.0
Logical validity	53.1	56.2
No redundancy	42.2	49.4
Overall preference	51.0	55.8
`gemini-3.8-flash`		
Problem groundedness	51.8	53.4
Step coherence	49.7	57.8
Logical validity	50.0	55.8
No redundancy	43.0	52.2
Overall preference	46.1	55.0

D EXPERIENTIAL SELF-TEACHING

Setup. Self-teaching replaces the larger teacher with the student itself, given privileged guidance as additional context: the frozen Qwen3-0.6B student proposes with the guidance placed before the question in its prompt, while the gate and fallback policy use the same checkpoint with the question alone. Following LLM-as-a-Coach (Ye et al., 2026), we build the guidance in three steps. An external LLM, `gpt-5.6-luna`, first writes a question-specific rubric from the question alone. The student then generates rollouts, and our verifier checks their correctness. Finally, the same LLM acts as a coach: from the question, rubric, rollouts, and correctness bits, it writes guidance supplying an approach, intermediate relations, corrections, and checks, averaging 313.8 tokens. The coach never receives the reference answer or solution; notes are audited for answer leakage and used only during generation, never as SFT inputs or targets.

Self-Teacher RFT samples from the guided proposer without gating; Self-Teacher IPD uses teacher-propose gating at $\tau = 0.1\%$. All 1,000 s1K questions receive up to eight attempts, retaining an accepted correct trace or a partial trace of at most 128 tokens, and SFT uses 15 epochs at a learning rate of 3×10^{-5}. The four-benchmark comparison uses one base seed, with $k = 8$ for AIME and $k = 4$ for GPQA-D and MATH500; Average is the unweighted mean of the four benchmark means.

Table 18: Four-benchmark results for guided self-teaching with Qwen3-0.6B on s1K. Error terms give the standard deviation across evaluation attempts from one evaluation seed.

Method	AIME24	AIME25	GPQA-D	MATH500	Average
Vanilla	7.92 ± 4.34	16.67 ± 3.56	22.10 ± 2.35	71.30 ± 1.58	29.50
Student RFT	4.58 ± 3.05	13.75 ± 3.30	26.01 ± 1.87	72.50 ± 0.35	29.21
Self-Teacher RFT	5.83 ± 3.88	12.50 ± 3.88	20.20 ± 0.71	68.70 ± 1.80	26.81
Self-Teacher IPD (0.1%)	4.17 ± 4.27	14.17 ± 4.96	25.63 ± 2.98	70.30 ± 0.26	28.57
Teacher RFT (8B)	9.58 ± 2.14	20.42 ± 4.86	28.03 ± 3.51	74.25 ± 0.57	33.07
IPD, T-P (8B, 0.1%)	13.75 ± 1.18	20.83 ± 3.88	32.70 ± 2.45	77.10 ± 1.32	36.10

Results. Table 18 reports the four original benchmarks for this experiment, separately from the headline seven-benchmark aggregates. Self-Teacher IPD produces 248 correct full traces, compared with 219 for Student RFT and 206 for Self-Teacher RFT. Guidance also makes the traces more surprising to the student: mean training-target surprisal under the frozen student rises from 0.2366 for Student RFT to 0.4353 for Self-Teacher RFT and 0.5312 for Self-Teacher IPD. Yet the trained student is no better: Self-Teacher IPD averages 28.57, below both Student RFT (29.21) and Vanilla (29.50), and Self-Teacher RFT is lower still at 26.81. The traces still contain only reasoning the student itself produces, consistent with Sections 5.1 and 5.2: compatibility without a stronger source of reasoning adds little.

E Full Benchmark Results

E.1 Results Across Training Seeds

For fine-tuned models, seeds 42, 1, and 2026 index three independent SFT training runs of the same condition on the same data; they are not evaluation sampling seeds. The subscripts in Tables 19–22 instead give the dispersion across the k matched attempt runs within one evaluation, and the across-seed standard deviation for fine-tuned models in Table 1 measures how much the training seed moves the result.

Tables 19–22 show that the strongest seven-benchmark average belongs to a T-P configuration in all twelve setting and training-seed pairs. T-P at $\tau = 1\%$ exceeds Teacher RFT in all twelve pairs, and at the representative $\tau = 0.1\%$ in eleven; the exception is EXAONE at seed 1, where it trails by 0.43 points.

Table 19: Accuracy by seed for Qwen3-0.6B/8B on s1K. For fine-tuned models, seeds identify independent training runs; subscripts give the within-run standard deviation over evaluation attempts.

Method	Seed	AIME24	AIME25	AIME26	HMMT26	MATH500	Olympiad	GPQA-D	Average
Vanilla	42	$7.92_{\pm 4.34}$	$16.67_{\pm 3.56}$	$9.58_{\pm 4.15}$	$9.85_{\pm 5.06}$	$71.30_{\pm 1.58}$	$38.06_{\pm 1.45}$	$22.10_{\pm 2.35}$	25.07
Vanilla	1	$6.25_{\pm 4.15}$	$15.42_{\pm 3.54}$	$11.67_{\pm 3.09}$	$8.71_{\pm 2.53}$	$71.30_{\pm 1.91}$	$37.54_{\pm 1.38}$	$23.36_{\pm 1.12}$	24.89
Vanilla	2026	$4.58_{\pm 5.02}$	$12.92_{\pm 2.14}$	$10.00_{\pm 5.63}$	$8.33_{\pm 3.88}$	$70.60_{\pm 0.95}$	$37.87_{\pm 1.73}$	$21.59_{\pm 3.43}$	23.70
Direct Distillation	42	$5.00_{\pm 3.56}$	$19.17_{\pm 2.95}$	$10.83_{\pm 2.95}$	$7.58_{\pm 2.81}$	$71.20_{\pm 0.97}$	$35.65_{\pm 1.56}$	$19.82_{\pm 1.51}$	24.18
Direct Distillation	1	$5.00_{\pm 3.09}$	$15.83_{\pm 2.95}$	$10.00_{\pm 5.35}$	$9.85_{\pm 4.21}$	$71.20_{\pm 1.10}$	$36.16_{\pm 1.10}$	$19.70_{\pm 1.80}$	23.96
Direct Distillation	2026	$4.58_{\pm 3.05}$	$20.00_{\pm 4.36}$	$9.58_{\pm 6.53}$	$7.95_{\pm 3.21}$	$72.50_{\pm 1.89}$	$35.79_{\pm 0.78}$	$19.82_{\pm 2.75}$	24.32
Student RFT	42	$4.58_{\pm 3.05}$	$13.75_{\pm 3.30}$	$10.00_{\pm 3.98}$	$8.71_{\pm 3.78}$	$72.50_{\pm 0.35}$	$38.02_{\pm 1.07}$	$26.01_{\pm 1.87}$	24.80
Student RFT	1	$7.08_{\pm 2.78}$	$15.00_{\pm 3.56}$	$10.83_{\pm 4.96}$	$7.58_{\pm 2.29}$	$69.90_{\pm 1.41}$	$38.54_{\pm 1.34}$	$25.63_{\pm 1.91}$	24.94
Student RFT	2026	$6.67_{\pm 6.17}$	$14.58_{\pm 4.69}$	$10.00_{\pm 5.04}$	$6.82_{\pm 4.51}$	$71.85_{\pm 1.00}$	$38.17_{\pm 0.96}$	$27.65_{\pm 6.20}$	25.11
Teacher RFT	42	$9.58_{\pm 2.14}$	$20.42_{\pm 4.86}$	$10.00_{\pm 5.04}$	$10.98_{\pm 2.25}$	$74.25_{\pm 0.57}$	$41.10_{\pm 0.90}$	$28.03_{\pm 3.51}$	27.77
Teacher RFT	1	$5.83_{\pm 3.88}$	$20.00_{\pm 3.98}$	$8.75_{\pm 3.05}$	$14.39_{\pm 3.53}$	$74.85_{\pm 2.30}$	$41.25_{\pm 0.44}$	$26.77_{\pm 1.80}$	27.41
Teacher RFT	2026	$7.08_{\pm 3.30}$	$19.17_{\pm 1.54}$	$14.58_{\pm 4.69}$	$10.23_{\pm 3.95}$	$74.10_{\pm 0.50}$	$41.58_{\pm 0.99}$	$27.65_{\pm 2.81}$	27.77
IPD, S-P (1%)	42	$10.83_{\pm 4.63}$	$19.58_{\pm 4.52}$	$18.75_{\pm 5.33}$	$13.64_{\pm 3.62}$	$74.95_{\pm 0.68}$	$42.66_{\pm 1.07}$	$25.25_{\pm 3.22}$	29.38
IPD, S-P (1%)	1	$11.25_{\pm 2.48}$	$19.58_{\pm 3.30}$	$13.33_{\pm 4.36}$	$9.85_{\pm 2.14}$	$78.05_{\pm 1.34}$	$43.32_{\pm 0.79}$	$23.61_{\pm 1.77}$	28.43
IPD, S-P (1%)	2026	$11.67_{\pm 1.78}$	$19.58_{\pm 4.15}$	$16.25_{\pm 7.22}$	$13.64_{\pm 3.97}$	$76.35_{\pm 0.90}$	$43.40_{\pm 1.13}$	$25.51_{\pm 1.73}$	29.48
IPD, S-P (0.1%)	42	$7.92_{\pm 3.05}$	$20.00_{\pm 5.63}$	$14.58_{\pm 6.16}$	$12.50_{\pm 4.42}$	$76.10_{\pm 1.32}$	$42.14_{\pm 0.63}$	$24.12_{\pm 1.12}$	28.19
IPD, S-P (0.1%)	1	$7.50_{\pm 3.45}$	$19.58_{\pm 4.15}$	$13.33_{\pm 3.98}$	$13.26_{\pm 3.60}$	$75.00_{\pm 0.97}$	$41.84_{\pm 1.16}$	$25.00_{\pm 2.42}$	27.93
IPD, S-P (0.1%)	2026	$12.92_{\pm 7.65}$	$20.00_{\pm 4.36}$	$14.58_{\pm 5.62}$	$13.26_{\pm 6.26}$	$76.95_{\pm 2.41}$	$40.80_{\pm 1.28}$	$23.74_{\pm 2.51}$	28.89
IPD, S-P (0.01%)	42	$9.17_{\pm 2.95}$	$17.08_{\pm 4.86}$	$16.25_{\pm 3.30}$	$11.36_{\pm 3.88}$	$75.25_{\pm 1.51}$	$42.54_{\pm 0.78}$	$29.17_{\pm 1.26}$	28.69
IPD, S-P (0.01%)	1	$8.33_{\pm 5.35}$	$18.75_{\pm 3.96}$	$13.75_{\pm 4.86}$	$9.85_{\pm 2.14}$	$74.25_{\pm 0.82}$	$41.43_{\pm 1.47}$	$32.32_{\pm 2.51}$	28.38
IPD, S-P (0.01%)	2026	$10.42_{\pm 4.15}$	$20.00_{\pm 3.56}$	$16.67_{\pm 3.98}$	$11.74_{\pm 3.41}$	$74.80_{\pm 0.71}$	$41.62_{\pm 2.18}$	$28.66_{\pm 2.31}$	29.13
IPD, T-P (1%)	42	$11.25_{\pm 6.16}$	$20.42_{\pm 2.78}$	$12.50_{\pm 2.36}$	$12.50_{\pm 4.11}$	$77.75_{\pm 0.25}$	$43.10_{\pm 0.61}$	$31.69_{\pm 1.82}$	29.89
IPD, T-P (1%)	1	$7.50_{\pm 3.88}$	$22.08_{\pm 5.62}$	$12.92_{\pm 3.30}$	$12.88_{\pm 4.79}$	$78.15_{\pm 2.36}$	$43.92_{\pm 1.04}$	$29.17_{\pm 2.52}$	29.52
IPD, T-P (1%)	2026	$12.08_{\pm 5.89}$	$17.92_{\pm 3.96}$	$10.83_{\pm 4.27}$	$15.53_{\pm 3.78}$	$77.45_{\pm 0.66}$	$42.51_{\pm 0.55}$	$29.04_{\pm 1.34}$	29.34
IPD, T-P (0.1%)	42	$13.75_{\pm 1.18}$	$20.83_{\pm 3.88}$	$17.08_{\pm 4.52}$	$11.74_{\pm 4.42}$	$77.10_{\pm 1.32}$	$43.84_{\pm 1.11}$	$32.70_{\pm 2.45}$	31.01
IPD, T-P (0.1%)	1	$7.92_{\pm 5.02}$	$20.00_{\pm 4.71}$	$11.67_{\pm 5.63}$	$11.74_{\pm 6.96}$	$75.10_{\pm 1.14}$	$42.21_{\pm 0.71}$	$32.45_{\pm 0.64}$	28.73
IPD, T-P (0.1%)	2026	$8.75_{\pm 2.48}$	$20.83_{\pm 4.27}$	$13.33_{\pm 4.71}$	$13.64_{\pm 3.62}$	$76.70_{\pm 1.31}$	$42.62_{\pm 1.52}$	$30.18_{\pm 4.56}$	29.44
IPD, T-P (0.01%)	42	$8.33_{\pm 3.56}$	$16.67_{\pm 4.71}$	$13.33_{\pm 3.09}$	$12.88_{\pm 3.14}$	$75.95_{\pm 0.91}$	$41.36_{\pm 1.17}$	$35.35_{\pm 1.49}$	29.12
IPD, T-P (0.01%)	1	$9.17_{\pm 4.96}$	$20.42_{\pm 3.30}$	$14.58_{\pm 4.34}$	$17.42_{\pm 3.14}$	$74.95_{\pm 0.53}$	$41.36_{\pm 1.08}$	$37.88_{\pm 3.57}$	30.83
IPD, T-P (0.01%)	2026	$7.08_{\pm 4.52}$	$22.50_{\pm 3.88}$	$11.67_{\pm 4.36}$	$12.12_{\pm 4.29}$	$75.40_{\pm 1.19}$	$41.51_{\pm 1.02}$	$38.26_{\pm 2.75}$	29.79

Table 20: Accuracy by seed for EXAONE 4.0 1.2B/32B on s1K. For fine-tuned models, seeds identify independent training runs; subscripts give the within-run standard deviation over evaluation attempts.

Method	Seed	AIME24	AIME25	AIME26	HMMT26	MATH500	Olympiad	GPQA-D	Average
Vanilla	42	$40.00_{\pm 5.04}$	$37.08_{\pm 2.14}$	$49.58_{\pm 7.86}$	$28.79_{\pm 5.37}$	$89.60_{\pm 1.31}$	$65.91_{\pm 1.16}$	$46.34_{\pm 1.12}$	51.04
Vanilla	1	$37.92_{\pm 4.34}$	$42.92_{\pm 3.75}$	$46.67_{\pm 3.56}$	$33.33_{\pm 4.58}$	$89.85_{\pm 0.66}$	$67.21_{\pm 0.71}$	$48.61_{\pm 1.45}$	52.36
Vanilla	2026	$37.50_{\pm 2.95}$	$43.33_{\pm 5.91}$	$52.50_{\pm 4.63}$	$28.41_{\pm 5.36}$	$90.00_{\pm 1.42}$	$64.91_{\pm 1.32}$	$48.11_{\pm 1.95}$	52.11
Direct Distillation	42	$31.67_{\pm 5.91}$	$39.17_{\pm 3.88}$	$44.17_{\pm 5.27}$	$32.20_{\pm 2.78}$	$89.40_{\pm 0.78}$	$58.90_{\pm 1.39}$	$33.96_{\pm 1.62}$	47.07
Direct Distillation	1	$32.92_{\pm 4.86}$	$36.25_{\pm 5.47}$	$44.17_{\pm 4.96}$	$30.68_{\pm 4.42}$	$89.90_{\pm 0.48}$	$58.79_{\pm 0.53}$	$33.21_{\pm 3.31}$	46.56
Direct Distillation	2026	$35.00_{\pm 6.67}$	$35.42_{\pm 6.65}$	$43.75_{\pm 4.86}$	$32.20_{\pm 3.60}$	$90.00_{\pm 0.95}$	$58.90_{\pm 0.62}$	$34.60_{\pm 1.68}$	47.12
Student RFT	42	$39.17_{\pm 5.27}$	$46.25_{\pm 5.18}$	$53.33_{\pm 7.56}$	$33.33_{\pm 2.81}$	$90.40_{\pm 0.71}$	$67.69_{\pm 0.61}$	$47.10_{\pm 2.42}$	53.90
Student RFT	1	$42.08_{\pm 4.69}$	$44.17_{\pm 6.36}$	$54.17_{\pm 6.11}$	$32.58_{\pm 3.88}$	$90.10_{\pm 1.59}$	$66.95_{\pm 1.20}$	$47.98_{\pm 4.76}$	54.00
Student RFT	2026	$40.00_{\pm 5.91}$	$42.92_{\pm 4.52}$	$53.75_{\pm 6.53}$	$32.95_{\pm 4.11}$	$90.95_{\pm 0.66}$	$68.40_{\pm 1.28}$	$51.39_{\pm 2.42}$	54.34
Teacher RFT	42	$48.75_{\pm 5.62}$	$53.33_{\pm 6.42}$	$57.92_{\pm 7.96}$	$35.61_{\pm 4.79}$	$92.40_{\pm 1.08}$	$67.84_{\pm 1.85}$	$50.13_{\pm 1.95}$	58.00
Teacher RFT	1	$48.75_{\pm 4.69}$	$55.83_{\pm 7.07}$	$57.50_{\pm 7.07}$	$37.50_{\pm 5.36}$	$90.65_{\pm 1.18}$	$67.84_{\pm 2.27}$	$49.87_{\pm 3.63}$	58.28
Teacher RFT	2026	$47.92_{\pm 5.33}$	$52.50_{\pm 6.36}$	$56.67_{\pm 6.90}$	$35.23_{\pm 4.27}$	$90.85_{\pm 0.57}$	$68.92_{\pm 0.61}$	$50.63_{\pm 1.99}$	57.53
IPD, S-P (1%)	42	$43.75_{\pm 5.18}$	$51.67_{\pm 7.13}$	$58.75_{\pm 7.33}$	$32.58_{\pm 4.79}$	$91.65_{\pm 0.72}$	$69.62_{\pm 0.14}$	$48.61_{\pm 3.15}$	56.66
IPD, S-P (1%)	1	$42.08_{\pm 7.11}$	$48.75_{\pm 7.33}$	$57.50_{\pm 11.23}$	$33.71_{\pm 4.42}$	$91.25_{\pm 0.53}$	$68.40_{\pm 0.76}$	$51.14_{\pm 2.20}$	56.12
IPD, S-P (1%)	2026	$46.67_{\pm 9.59}$	$52.92_{\pm 5.47}$	$54.17_{\pm 6.36}$	$33.71_{\pm 3.00}$	$90.30_{\pm 0.93}$	$68.77_{\pm 1.28}$	$48.99_{\pm 1.70}$	56.50
IPD, S-P (0.1%)	42	$44.17_{\pm 3.45}$	$47.92_{\pm 5.33}$	$53.33_{\pm 6.90}$	$34.85_{\pm 5.84}$	$90.95_{\pm 0.10}$	$67.88_{\pm 1.65}$	$50.25_{\pm 2.90}$	55.62
IPD, S-P (0.1%)	1	$45.42_{\pm 3.05}$	$47.92_{\pm 7.75}$	$55.83_{\pm 4.96}$	$32.20_{\pm 4.27}$	$90.85_{\pm 0.66}$	$69.44_{\pm 1.52}$	$49.75_{\pm 1.34}$	55.91
IPD, S-P (0.1%)	2026	$45.83_{\pm 7.51}$	$46.67_{\pm 7.13}$	$56.67_{\pm 4.36}$	$34.09_{\pm 6.22}$	$90.45_{\pm 0.66}$	$68.32_{\pm 0.86}$	$51.39_{\pm 1.26}$	56.20
IPD, S-P (0.01%)	42	$45.83_{\pm 4.96}$	$50.00_{\pm 7.56}$	$53.75_{\pm 5.47}$	$35.61_{\pm 8.83}$	$90.70_{\pm 0.62}$	$68.03_{\pm 1.05}$	$50.00_{\pm 1.43}$	56.27
IPD, S-P (0.01%)	1	$41.25_{\pm 5.33}$	$49.17_{\pm 6.61}$	$56.67_{\pm 5.91}$	$34.85_{\pm 4.29}$	$89.00_{\pm 0.16}$	$69.51_{\pm 0.91}$	$50.88_{\pm 2.20}$	55.90
IPD, S-P (0.01%)	2026	$42.50_{\pm 6.11}$	$47.92_{\pm 10.07}$	$57.08_{\pm 3.30}$	$34.09_{\pm 6.01}$	$90.10_{\pm 0.62}$	$68.66_{\pm 1.75}$	$51.89_{\pm 3.51}$	56.03
IPD, T-P (1%)	42	$47.08_{\pm 5.18}$	$55.00_{\pm 4.36}$	$59.58_{\pm 4.86}$	$33.71_{\pm 3.41}$	$92.20_{\pm 0.99}$	$69.03_{\pm 0.93}$	$49.49_{\pm 2.06}$	58.01
IPD, T-P (1%)	1	$49.17_{\pm 6.11}$	$58.33_{\pm 8.36}$	$62.50_{\pm 4.27}$	$34.85_{\pm 4.86}$	$91.55_{\pm 0.25}$	$69.44_{\pm 0.93}$	$53.79_{\pm 3.04}$	59.95
IPD, T-P (1%)	2026	$47.50_{\pm 2.95}$	$57.08_{\pm 3.75}$	$60.42_{\pm 3.30}$	$36.74_{\pm 5.94}$	$91.50_{\pm 0.48}$	$70.25_{\pm 0.15}$	$50.76_{\pm 3.41}$	59.18
IPD, T-P (0.1%)	42	$46.25_{\pm 3.30}$	$55.83_{\pm 2.95}$	$64.17_{\pm 6.11}$	$34.09_{\pm 5.78}$	$92.15_{\pm 1.22}$	$67.69_{\pm 1.71}$	$53.16_{\pm 3.94}$	59.05
IPD, T-P (0.1%)	1	$46.67_{\pm 2.52}$	$57.08_{\pm 5.47}$	$59.17_{\pm 6.61}$	$31.82_{\pm 4.29}$	$91.40_{\pm 0.57}$	$68.03_{\pm 0.89}$	$50.76_{\pm 1.96}$	57.85
IPD, T-P (0.1%)	2026	$47.50_{\pm 4.96}$	$50.83_{\pm 5.27}$	$60.83_{\pm 5.56}$	$32.58_{\pm 6.22}$	$91.75_{\pm 0.75}$	$67.51_{\pm 0.27}$	$52.02_{\pm 4.31}$	57.57
IPD, T-P (0.01%)	42	$49.17_{\pm 3.88}$	$56.67_{\pm 5.35}$	$57.50_{\pm 7.29}$	$36.36_{\pm 3.97}$	$91.45_{\pm 0.81}$	$68.95_{\pm 1.07}$	$45.08_{\pm 2.81}$	57.88
IPD, T-P (0.01%)	1	$45.42_{\pm 3.96}$	$57.08_{\pm 6.77}$	$65.42_{\pm 5.33}$	$29.55_{\pm 4.79}$	$91.55_{\pm 0.44}$	$69.47_{\pm 1.08}$	$47.10_{\pm 3.07}$	57.94
IPD, T-P (0.01%)	2026	$45.83_{\pm 5.84}$	$58.33_{\pm 6.17}$	$60.00_{\pm 8.54}$	$32.95_{\pm 5.23}$	$91.15_{\pm 0.75}$	$70.18_{\pm 0.74}$	$47.98_{\pm 3.14}$	58.06

Table 21: Accuracy by seed for Qwen3-0.6B/8B on LIMO. For fine-tuned models, seeds identify independent training runs; subscripts give the within-run standard deviation over evaluation attempts.

Method	Seed	AIME24	AIME25	AIME26	HMMT26	MATH500	Olympiad	GPQA-D	Average
Vanilla	42	$7.92_{\pm 4.34}$	$16.67_{\pm 3.56}$	$9.58_{\pm 4.15}$	$9.85_{\pm 5.06}$	$71.30_{\pm 1.58}$	$38.06_{\pm 1.45}$	$22.10_{\pm 2.35}$	25.07
Vanilla	1	$6.25_{\pm 4.15}$	$15.42_{\pm 3.54}$	$11.67_{\pm 3.09}$	$8.71_{\pm 2.53}$	$71.30_{\pm 1.91}$	$37.54_{\pm 1.38}$	$23.36_{\pm 1.12}$	24.89
Vanilla	2026	$4.58_{\pm 5.02}$	$12.92_{\pm 2.14}$	$10.00_{\pm 5.63}$	$8.33_{\pm 3.88}$	$70.60_{\pm 0.95}$	$37.87_{\pm 1.73}$	$21.59_{\pm 3.43}$	23.70
Direct Distillation	42	$9.17_{\pm 7.72}$	$17.08_{\pm 5.18}$	$10.83_{\pm 4.27}$	$7.95_{\pm 3.21}$	$70.85_{\pm 1.95}$	$38.24_{\pm 0.44}$	$17.68_{\pm 0.71}$	24.54
Direct Distillation	1	$7.50_{\pm 3.45}$	$15.00_{\pm 5.35}$	$9.17_{\pm 4.27}$	$8.71_{\pm 1.94}$	$70.95_{\pm 0.60}$	$38.24_{\pm 1.21}$	$17.30_{\pm 2.65}$	23.84
Direct Distillation	2026	$7.92_{\pm 3.05}$	$17.50_{\pm 4.96}$	$10.83_{\pm 3.88}$	$7.95_{\pm 3.95}$	$71.95_{\pm 0.57}$	$39.17_{\pm 1.23}$	$17.17_{\pm 2.10}$	24.64
Student RFT	42	$2.08_{\pm 2.48}$	$17.08_{\pm 4.15}$	$7.92_{\pm 2.48}$	$8.71_{\pm 2.53}$	$71.70_{\pm 1.87}$	$37.05_{\pm 1.45}$	$24.49_{\pm 1.91}$	24.15
Student RFT	1	$3.75_{\pm 5.76}$	$15.00_{\pm 6.42}$	$9.58_{\pm 3.30}$	$9.85_{\pm 4.21}$	$67.50_{\pm 1.18}$	$37.69_{\pm 1.55}$	$23.36_{\pm 1.51}$	23.82
Student RFT	2026	$7.08_{\pm 3.30}$	$13.33_{\pm 2.52}$	$11.25_{\pm 3.54}$	$7.58_{\pm 2.81}$	$68.00_{\pm 1.45}$	$37.76_{\pm 1.46}$	$24.49_{\pm 0.97}$	24.21
Teacher RFT	42	$7.92_{\pm 3.54}$	$17.92_{\pm 4.34}$	$15.00_{\pm 3.09}$	$11.36_{\pm 3.88}$	$74.70_{\pm 0.58}$	$41.06_{\pm 1.21}$	$28.41_{\pm 2.20}$	28.05
Teacher RFT	1	$7.50_{\pm 5.27}$	$17.08_{\pm 2.14}$	$16.25_{\pm 6.03}$	$12.12_{\pm 4.29}$	$75.15_{\pm 1.05}$	$41.73_{\pm 1.68}$	$22.73_{\pm 2.14}$	27.51
Teacher RFT	2026	$7.92_{\pm 2.48}$	$14.17_{\pm 4.96}$	$17.08_{\pm 3.30}$	$12.12_{\pm 1.62}$	$73.45_{\pm 1.18}$	$40.91_{\pm 1.77}$	$23.74_{\pm 2.47}$	27.06
IPD, S-P (1%)	42	$7.92_{\pm 4.69}$	$18.75_{\pm 2.48}$	$13.75_{\pm 7.00}$	$14.02_{\pm 3.21}$	$77.40_{\pm 2.73}$	$42.99_{\pm 0.95}$	$25.38_{\pm 3.23}$	28.60
IPD, S-P (1%)	1	$10.00_{\pm 3.56}$	$18.75_{\pm 5.89}$	$15.83_{\pm 3.88}$	$14.77_{\pm 1.94}$	$75.95_{\pm 1.94}$	$43.51_{\pm 0.85}$	$28.79_{\pm 2.70}$	29.66
IPD, S-P (1%)	2026	$7.08_{\pm 3.30}$	$18.75_{\pm 4.34}$	$14.17_{\pm 3.45}$	$15.15_{\pm 2.29}$	$76.45_{\pm 0.66}$	$42.95_{\pm 0.83}$	$26.77_{\pm 2.86}$	28.76
IPD, S-P (0.1%)	42	$8.33_{\pm 3.56}$	$18.33_{\pm 3.98}$	$13.33_{\pm 3.98}$	$11.74_{\pm 4.71}$	$75.25_{\pm 1.89}$	$41.77_{\pm 1.54}$	$26.39_{\pm 1.91}$	27.88
IPD, S-P (0.1%)	1	$7.50_{\pm 3.45}$	$19.17_{\pm 6.84}$	$12.08_{\pm 3.96}$	$10.98_{\pm 4.84}$	$75.70_{\pm 0.48}$	$41.51_{\pm 0.90}$	$29.67_{\pm 1.99}$	28.09
IPD, S-P (0.1%)	2026	$7.92_{\pm 3.96}$	$19.17_{\pm 3.45}$	$13.33_{\pm 2.52}$	$12.50_{\pm 6.57}$	$75.80_{\pm 0.82}$	$41.14_{\pm 0.87}$	$26.39_{\pm 4.07}$	28.03
IPD, S-P (0.01%)	42	$9.58_{\pm 4.15}$	$17.08_{\pm 5.76}$	$14.17_{\pm 6.11}$	$12.50_{\pm 3.00}$	$75.20_{\pm 0.85}$	$41.77_{\pm 0.83}$	$24.49_{\pm 2.66}$	27.83
IPD, S-P (0.01%)	1	$8.33_{\pm 3.98}$	$15.00_{\pm 5.35}$	$13.33_{\pm 5.04}$	$12.12_{\pm 2.81}$	$75.40_{\pm 1.19}$	$40.91_{\pm 1.74}$	$25.88_{\pm 2.78}$	27.28
IPD, S-P (0.01%)	2026	$9.17_{\pm 2.95}$	$16.25_{\pm 2.14}$	$16.25_{\pm 3.30}$	$9.09_{\pm 3.24}$	$74.60_{\pm 1.98}$	$40.91_{\pm 0.37}$	$23.99_{\pm 3.34}$	27.18
IPD, T-P (1%)	42	$11.67_{\pm 4.71}$	$18.33_{\pm 6.42}$	$16.67_{\pm 3.09}$	$12.88_{\pm 3.53}$	$78.05_{\pm 1.43}$	$45.10_{\pm 0.54}$	$28.41_{\pm 2.27}$	30.16
IPD, T-P (1%)	1	$9.17_{\pm 4.63}$	$20.00_{\pm 5.04}$	$13.75_{\pm 6.53}$	$12.50_{\pm 4.11}$	$78.35_{\pm 0.50}$	$45.10_{\pm 1.02}$	$27.78_{\pm 1.70}$	29.52
IPD, T-P (1%)	2026	$7.92_{\pm 3.96}$	$19.58_{\pm 3.30}$	$12.50_{\pm 2.36}$	$12.88_{\pm 2.69}$	$77.55_{\pm 1.48}$	$44.99_{\pm 0.95}$	$29.92_{\pm 1.91}$	29.34
IPD, T-P (0.1%)	42	$9.17_{\pm 2.95}$	$17.92_{\pm 4.69}$	$17.50_{\pm 5.56}$	$13.64_{\pm 3.24}$	$76.15_{\pm 1.18}$	$44.73_{\pm 1.60}$	$31.44_{\pm 2.20}$	30.08
IPD, T-P (0.1%)	1	$10.42_{\pm 4.52}$	$18.33_{\pm 4.36}$	$15.42_{\pm 3.05}$	$11.74_{\pm 5.23}$	$76.75_{\pm 0.93}$	$43.99_{\pm 1.50}$	$31.57_{\pm 2.81}$	29.75
IPD, T-P (0.1%)	2026	$8.33_{\pm 6.67}$	$17.50_{\pm 4.27}$	$15.83_{\pm 5.84}$	$13.26_{\pm 4.56}$	$77.45_{\pm 0.38}$	$43.55_{\pm 1.50}$	$35.23_{\pm 1.56}$	30.16
IPD, T-P (0.01%)	42	$10.00_{\pm 2.52}$	$19.17_{\pm 7.07}$	$15.00_{\pm 5.35}$	$14.02_{\pm 3.95}$	$77.00_{\pm 1.55}$	$43.18_{\pm 1.10}$	$28.41_{\pm 2.04}$	29.54
IPD, T-P (0.01%)	1	$11.67_{\pm 5.63}$	$20.42_{\pm 4.15}$	$16.25_{\pm 3.75}$	$13.64_{\pm 2.29}$	$77.75_{\pm 1.29}$	$44.07_{\pm 1.95}$	$27.78_{\pm 0.82}$	30.22
IPD, T-P (0.01%)	2026	$12.92_{\pm 4.15}$	$18.33_{\pm 5.35}$	$13.75_{\pm 4.52}$	$12.88_{\pm 5.55}$	$76.90_{\pm 0.93}$	$44.14_{\pm 0.37}$	$27.90_{\pm 1.33}$	29.55

Table 22: Accuracy by seed for Qwen3-1.7B/8B on s1K. For fine-tuned models, seeds identify independent training runs; subscripts give the within-run standard deviation over evaluation attempts.

Method	Seed	AIME24	AIME25	AIME26	HMMT26	MATH500	Olympiad	GPQA-D	Average
Vanilla	42	$34.17_{\pm 6.11}$	$31.67_{\pm 3.56}$	$32.50_{\pm 5.27}$	$24.24_{\pm 4.29}$	$88.70_{\pm 0.74}$	$56.34_{\pm 0.52}$	$41.41_{\pm 2.70}$	44.15
Vanilla	1	$31.67_{\pm 4.71}$	$34.58_{\pm 5.33}$	$36.25_{\pm 7.00}$	$25.00_{\pm 2.14}$	$89.00_{\pm 0.59}$	$58.27_{\pm 1.03}$	$41.79_{\pm 3.51}$	45.22
Vanilla	2026	$29.58_{\pm 4.52}$	$31.67_{\pm 3.56}$	$31.67_{\pm 4.71}$	$26.89_{\pm 3.00}$	$88.45_{\pm 0.68}$	$56.82_{\pm 1.55}$	$43.43_{\pm 4.06}$	44.07
Direct Distillation	42	$25.42_{\pm 4.34}$	$26.25_{\pm 5.18}$	$25.00_{\pm 4.36}$	$25.00_{\pm 4.51}$	$87.00_{\pm 0.67}$	$54.04_{\pm 0.75}$	$24.49_{\pm 2.56}$	38.17
Direct Distillation	1	$23.75_{\pm 6.77}$	$26.25_{\pm 4.86}$	$27.92_{\pm 4.34}$	$24.24_{\pm 3.97}$	$87.10_{\pm 0.96}$	$54.30_{\pm 0.56}$	$27.40_{\pm 2.31}$	38.71
Direct Distillation	2026	$25.42_{\pm 5.02}$	$27.50_{\pm 4.63}$	$30.00_{\pm 4.71}$	$21.97_{\pm 3.14}$	$87.20_{\pm 0.49}$	$54.15_{\pm 0.77}$	$26.14_{\pm 1.33}$	38.91
Student RFT	42	$31.25_{\pm 5.62}$	$37.92_{\pm 2.48}$	$37.50_{\pm 3.45}$	$26.14_{\pm 2.25}$	$87.90_{\pm 1.09}$	$57.31_{\pm 0.53}$	$39.52_{\pm 2.24}$	45.36
Student RFT	1	$35.00_{\pm 6.67}$	$33.75_{\pm 3.30}$	$35.00_{\pm 4.71}$	$28.03_{\pm 3.53}$	$88.30_{\pm 0.53}$	$57.46_{\pm 1.03}$	$38.26_{\pm 1.39}$	45.11
Student RFT	2026	$33.33_{\pm 6.42}$	$34.58_{\pm 7.33}$	$34.58_{\pm 5.02}$	$27.65_{\pm 2.53}$	$88.80_{\pm 1.43}$	$58.09_{\pm 0.15}$	$40.03_{\pm 2.59}$	45.29
Teacher RFT	42	$32.92_{\pm 4.15}$	$35.42_{\pm 3.05}$	$37.92_{\pm 9.42}$	$26.52_{\pm 3.53}$	$88.80_{\pm 0.69}$	$58.46_{\pm 0.72}$	$42.42_{\pm 1.30}$	46.07
Teacher RFT	1	$29.58_{\pm 4.86}$	$35.00_{\pm 5.04}$	$38.33_{\pm 6.90}$	$25.38_{\pm 5.11}$	$88.90_{\pm 0.89}$	$58.16_{\pm 1.73}$	$42.30_{\pm 2.35}$	45.38
Teacher RFT	2026	$30.42_{\pm 4.52}$	$34.58_{\pm 8.15}$	$41.67_{\pm 6.90}$	$28.79_{\pm 2.81}$	$89.15_{\pm 1.08}$	$57.68_{\pm 0.43}$	$41.41_{\pm 1.30}$	46.24
IPD, S-P (1%)	42	$33.75_{\pm 6.53}$	$35.42_{\pm 5.89}$	$40.00_{\pm 3.98}$	$30.30_{\pm 4.58}$	$89.35_{\pm 1.11}$	$59.68_{\pm 0.41}$	$41.29_{\pm 1.12}$	47.11
IPD, S-P (1%)	1	$34.17_{\pm 9.04}$	$35.42_{\pm 3.05}$	$39.58_{\pm 5.18}$	$25.76_{\pm 5.37}$	$89.60_{\pm 1.31}$	$58.53_{\pm 1.12}$	$41.79_{\pm 2.49}$	46.41
IPD, S-P (1%)	2026	$35.42_{\pm 6.65}$	$34.58_{\pm 4.34}$	$38.75_{\pm 7.75}$	$26.89_{\pm 4.42}$	$89.50_{\pm 0.76}$	$58.16_{\pm 0.90}$	$39.27_{\pm 1.33}$	46.08
IPD, S-P (0.1%)	42	$30.83_{\pm 6.11}$	$35.00_{\pm 1.78}$	$39.17_{\pm 5.27}$	$27.65_{\pm 4.11}$	$89.60_{\pm 0.97}$	$58.57_{\pm 0.82}$	$42.30_{\pm 1.82}$	46.16
IPD, S-P (0.1%)	1	$34.58_{\pm 3.96}$	$33.33_{\pm 4.71}$	$39.17_{\pm 7.29}$	$28.41_{\pm 3.60}$	$88.80_{\pm 1.28}$	$58.98_{\pm 0.75}$	$43.81_{\pm 1.62}$	46.73
IPD, S-P (0.1%)	2026	$36.67_{\pm 2.52}$	$31.67_{\pm 5.35}$	$43.33_{\pm 4.71}$	$28.41_{\pm 2.78}$	$89.75_{\pm 0.72}$	$59.35_{\pm 0.80}$	$42.55_{\pm 1.91}$	47.39
IPD, S-P (0.01%)	42	$33.33_{\pm 5.63}$	$34.58_{\pm 3.05}$	$39.17_{\pm 6.36}$	$29.17_{\pm 3.60}$	$88.70_{\pm 0.95}$	$58.57_{\pm 0.25}$	$41.16_{\pm 3.61}$	46.38
IPD, S-P (0.01%)	1	$35.83_{\pm 7.51}$	$33.33_{\pm 4.71}$	$33.33_{\pm 6.90}$	$30.30_{\pm 4.58}$	$89.35_{\pm 0.81}$	$59.12_{\pm 1.42}$	$44.32_{\pm 2.59}$	46.51
IPD, S-P (0.01%)	2026	$34.58_{\pm 8.15}$	$35.83_{\pm 4.96}$	$35.83_{\pm 4.96}$	$30.30_{\pm 3.97}$	$89.35_{\pm 1.04}$	$58.57_{\pm 0.82}$	$40.40_{\pm 2.40}$	46.41
IPD, T-P (1%)	42	$35.83_{\pm 7.07}$	$35.83_{\pm 6.84}$	$40.42_{\pm 5.47}$	$28.79_{\pm 4.58}$	$89.50_{\pm 0.58}$	$59.79_{\pm 0.85}$	$47.85_{\pm 1.91}$	48.29
IPD, T-P (1%)	1	$32.92_{\pm 4.86}$	$37.92_{\pm 3.54}$	$39.17_{\pm 6.36}$	$26.14_{\pm 3.60}$	$90.40_{\pm 0.43}$	$59.64_{\pm 0.79}$	$46.84_{\pm 1.62}$	47.57
IPD, T-P (1%)	2026	$32.08_{\pm 5.33}$	$37.50_{\pm 3.45}$	$41.25_{\pm 4.34}$	$29.17_{\pm 5.36}$	$90.10_{\pm 0.48}$	$59.27_{\pm 0.79}$	$46.46_{\pm 2.37}$	47.98
IPD, T-P (0.1%)	42	$37.92_{\pm 3.96}$	$39.58_{\pm 5.47}$	$42.08_{\pm 6.89}$	$32.20_{\pm 3.21}$	$89.55_{\pm 1.20}$	$59.53_{\pm 0.49}$	$44.70_{\pm 3.97}$	49.37
IPD, T-P (0.1%)	1	$34.58_{\pm 5.89}$	$35.00_{\pm 3.09}$	$39.17_{\pm 8.12}$	$28.41_{\pm 2.25}$	$90.45_{\pm 0.93}$	$60.27_{\pm 0.53}$	$43.56_{\pm 4.82}$	47.35
IPD, T-P (0.1%)	2026	$37.50_{\pm 5.56}$	$37.92_{\pm 9.75}$	$36.25_{\pm 3.30}$	$27.27_{\pm 3.62}$	$89.85_{\pm 1.11}$	$59.94_{\pm 0.53}$	$44.70_{\pm 0.65}$	47.63
IPD, T-P (0.01%)	42	$37.92_{\pm 5.89}$	$36.67_{\pm 6.17}$	$43.75_{\pm 7.22}$	$30.30_{\pm 3.97}$	$90.05_{\pm 0.87}$	$60.01_{\pm 0.66}$	$45.08_{\pm 2.16}$	49.11
IPD, T-P (0.01%)	1	$32.08_{\pm 8.72}$	$39.58_{\pm 3.30}$	$40.00_{\pm 7.13}$	$32.20_{\pm 3.60}$	$89.60_{\pm 0.95}$	$59.61_{\pm 1.14}$	$46.34_{\pm 2.87}$	48.49
IPD, T-P (0.01%)	2026	$33.33_{\pm 3.56}$	$35.00_{\pm 3.98}$	$41.67_{\pm 4.71}$	$32.20_{\pm 3.21}$	$89.70_{\pm 1.15}$	$59.79_{\pm 0.95}$	$44.44_{\pm 2.37}$	48.02

E.2 PROBLEM-SOLVING COVERAGE

Tables 23–26 report Pass@k for all four settings at the largest available budget: $k = 8$ for AIME and HMMT and $k = 4$ for MATH500, OlympiadBench, and GPQA-D. Results include per-seed coverage and its arithmetic mean; samples from different seeds are not pooled into a larger sampling budget. The tables support the 27-of-28 comparison in Section 5.4, including the two settings absent from Figure 5.

Figure 6 shows the full Pass@k curves for Qwen3-0.6B/8B on LIMO and Qwen3-1.7B/8B on s1K. IPD leads Vanilla and Direct Distillation on all fourteen benchmark–setting pairs at the largest matched sampling budget.

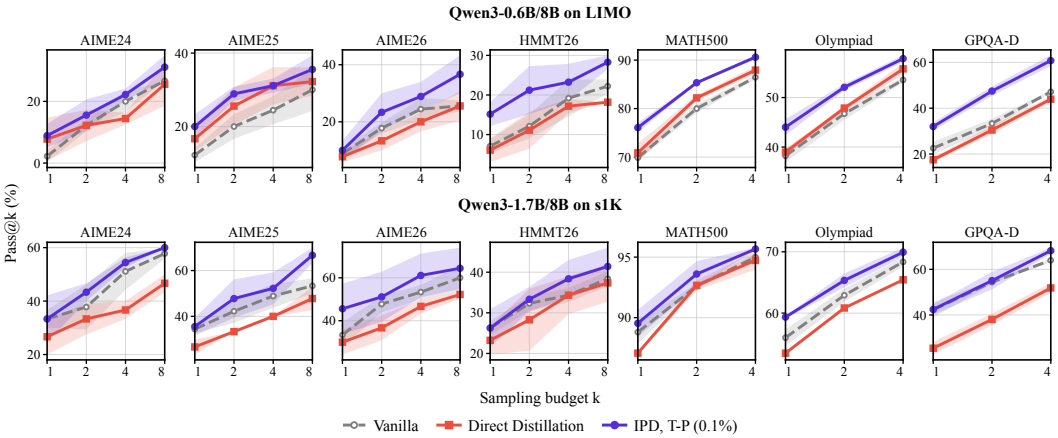


Figure 6: Pass@k for Qwen3-0.6B/8B on LIMO and Qwen3-1.7B/8B on s1K. Fine-tuned curves average three training seeds; bands show one sample standard deviation.

Table 23: Pass@8 on AIME and HMMT and Pass@4 on the other benchmarks for Qwen3-0.6B/8B on s1K. Mean rows average coverage across three seeds.

		Pass@8				Pass@4		
Method	**Seed**	**AIME24**	**AIME25**	**AIME26**	**HMMT26**	**MATH500**	**Olympiad**	**GPQA-D**
Vanilla	42	26.67	33.33	26.67	21.21	87.00	53.41	46.46
Vanilla	1	26.67	33.33	23.33	21.21	86.20	53.12	47.47
Vanilla	2026	26.67	23.33	26.67	24.24	86.20	54.15	47.47
Vanilla	*Mean*	26.67	30.00	25.56	22.22	86.47	53.56	47.14
Direct Distillation	42	16.67	30.00	23.33	27.27	86.40	50.30	47.98
Direct Distillation	1	20.00	33.33	26.67	24.24	87.00	52.08	45.45
Direct Distillation	2026	16.67	36.67	26.67	21.21	88.40	50.59	48.48
Direct Distillation	*Mean*	17.78	33.33	25.56	24.24	87.27	50.99	47.31
Student RFT	42	23.33	40.00	20.00	18.18	86.80	53.56	52.53
Student RFT	1	26.67	30.00	23.33	18.18	85.40	53.71	55.56
Student RFT	2026	30.00	33.33	23.33	21.21	87.00	52.37	57.07
Student RFT	*Mean*	26.67	34.44	22.22	19.19	86.40	53.21	55.05
Teacher RFT	42	33.33	40.00	26.67	30.30	88.00	56.08	52.53
Teacher RFT	1	23.33	30.00	16.67	27.27	89.80	56.23	57.58
Teacher RFT	2026	26.67	40.00	36.67	24.24	88.40	55.19	54.04
Teacher RFT	*Mean*	27.78	36.67	26.67	27.27	88.73	55.84	54.71
IPD, S-P (1%)	42	30.00	36.67	40.00	27.27	88.00	57.57	52.02
IPD, S-P (1%)	1	33.33	36.67	30.00	15.15	90.20	57.72	48.99
IPD, S-P (1%)	2026	30.00	40.00	43.33	30.30	90.40	56.97	50.51
IPD, S-P (1%)	*Mean*	31.11	37.78	37.78	24.24	89.53	57.42	50.51
IPD, S-P (0.1%)	42	26.67	46.67	30.00	27.27	90.60	55.49	48.99
IPD, S-P (0.1%)	1	23.33	36.67	23.33	24.24	90.00	55.93	47.98
IPD, S-P (0.1%)	2026	30.00	36.67	26.67	27.27	90.20	54.60	52.53
IPD, S-P (0.1%)	*Mean*	26.67	40.00	26.67	26.26	90.27	55.34	49.83
IPD, S-P (0.01%)	42	33.33	33.33	30.00	24.24	89.60	57.42	55.05
IPD, S-P (0.01%)	1	33.33	40.00	23.33	24.24	87.20	55.34	60.61
IPD, S-P (0.01%)	2026	30.00	40.00	36.67	27.27	89.20	56.53	55.56
IPD, S-P (0.01%)	*Mean*	32.22	37.78	30.00	25.25	88.67	56.43	57.07
IPD, T-P (1%)	42	33.33	43.33	30.00	24.24	90.20	58.61	67.17
IPD, T-P (1%)	1	30.00	46.67	40.00	27.27	90.60	58.16	57.07
IPD, T-P (1%)	2026	40.00	33.33	23.33	36.36	89.80	56.08	60.61
IPD, T-P (1%)	*Mean*	34.44	41.11	31.11	29.29	90.20	57.62	61.62
IPD, T-P (0.1%)	42	33.33	43.33	36.67	24.24	90.00	57.57	57.58
IPD, T-P (0.1%)	1	26.67	40.00	26.67	33.33	88.80	56.38	61.62
IPD, T-P (0.1%)	2026	30.00	40.00	33.33	33.33	89.80	56.53	59.60
IPD, T-P (0.1%)	*Mean*	30.00	41.11	32.22	30.30	89.53	56.82	59.60
IPD, T-P (0.01%)	42	33.33	30.00	33.33	30.30	88.80	56.53	62.63
IPD, T-P (0.01%)	1	33.33	36.67	30.00	39.39	88.60	54.90	69.19
IPD, T-P (0.01%)	2026	26.67	40.00	30.00	21.21	89.40	55.34	68.18
IPD, T-P (0.01%)	*Mean*	31.11	35.56	31.11	30.30	88.93	55.59	66.67

Table 24: Pass@8 on AIME and HMMT and Pass@4 on the other benchmarks for EXAONE 4.0 1.2B/32B on s1K. Mean rows average coverage across three seeds.

		Pass@8				Pass@4		
Method	**Seed**	**AIME24**	**AIME25**	**AIME26**	**HMMT26**	**MATH500**	**Olympiad**	**GPQA-D**
Vanilla	42	60.00	70.00	76.67	45.45	95.40	80.42	74.24
Vanilla	1	60.00	66.67	73.33	54.55	96.40	80.42	76.26
Vanilla	2026	63.33	70.00	80.00	51.52	95.60	79.08	73.74
Vanilla	*Mean*	61.11	68.89	76.67	50.51	95.80	79.97	74.75
Direct Distillation	42	56.67	73.33	73.33	39.39	96.00	70.62	59.09
Direct Distillation	1	60.00	60.00	70.00	39.39	96.00	69.88	57.58
Direct Distillation	2026	53.33	73.33	70.00	42.42	96.60	70.03	59.60
Direct Distillation	*Mean*	56.67	68.89	71.11	40.40	96.20	70.18	58.75
Student RFT	42	60.00	73.33	80.00	54.55	96.00	80.27	73.74
Student RFT	1	56.67	70.00	73.33	51.52	96.00	80.27	73.74
Student RFT	2026	63.33	73.33	76.67	54.55	96.60	80.86	76.77
Student RFT	*Mean*	60.00	72.22	76.67	53.54	96.20	80.46	74.75
Teacher RFT	42	73.33	76.67	83.33	60.61	97.60	79.53	76.26
Teacher RFT	1	66.67	83.33	80.00	60.61	97.00	81.60	76.77
Teacher RFT	2026	63.33	83.33	80.00	54.55	96.60	82.49	76.77
Teacher RFT	*Mean*	67.78	81.11	81.11	58.59	97.07	81.21	76.60
IPD, S-P (1%)	42	56.67	80.00	90.00	54.55	96.40	82.34	75.25
IPD, S-P (1%)	1	63.33	73.33	86.67	51.52	96.00	82.20	75.76
IPD, S-P (1%)	2026	70.00	83.33	86.67	54.55	95.80	81.31	74.24
IPD, S-P (1%)	*Mean*	63.33	78.89	87.78	53.54	96.07	81.95	75.08
IPD, S-P (0.1%)	42	73.33	73.33	83.33	57.58	97.40	81.16	72.22
IPD, S-P (0.1%)	1	63.33	80.00	80.00	48.48	96.40	83.09	76.26
IPD, S-P (0.1%)	2026	70.00	70.00	76.67	63.64	96.40	83.38	79.29
IPD, S-P (0.1%)	*Mean*	68.89	74.44	80.00	56.57	96.73	82.54	75.93
IPD, S-P (0.01%)	42	66.67	80.00	86.67	57.58	96.20	80.86	75.76
IPD, S-P (0.01%)	1	66.67	76.67	83.33	63.64	95.20	83.09	83.84
IPD, S-P (0.01%)	2026	63.33	76.67	80.00	60.61	95.80	81.31	78.28
IPD, S-P (0.01%)	*Mean*	65.56	77.78	83.33	60.61	95.73	81.75	79.29
IPD, T-P (1%)	42	63.33	80.00	83.33	54.55	97.00	82.79	75.25
IPD, T-P (1%)	1	66.67	80.00	86.67	54.55	96.20	82.05	81.82
IPD, T-P (1%)	2026	66.67	83.33	83.33	57.58	97.40	82.64	76.77
IPD, T-P (1%)	*Mean*	65.56	81.11	84.44	55.56	96.87	82.49	77.95
IPD, T-P (0.1%)	42	66.67	76.67	83.33	60.61	96.60	79.97	76.77
IPD, T-P (0.1%)	1	60.00	80.00	80.00	51.52	97.00	79.38	74.75
IPD, T-P (0.1%)	2026	66.67	73.33	83.33	60.61	96.60	79.82	77.27
IPD, T-P (0.1%)	*Mean*	64.44	76.67	82.22	57.58	96.73	79.72	76.26
IPD, T-P (0.01%)	42	66.67	76.67	83.33	54.55	96.20	82.05	73.23
IPD, T-P (0.01%)	1	60.00	80.00	83.33	48.48	97.40	83.23	73.23
IPD, T-P (0.01%)	2026	63.33	80.00	86.67	57.58	96.80	82.49	76.26
IPD, T-P (0.01%)	*Mean*	63.33	78.89	84.44	53.54	96.80	82.59	74.24

Table 25: Pass@8 on AIME and HMMT and Pass@4 on the other benchmarks for Qwen3-0.6B/8B on LIMO. Mean rows average coverage across three seeds.

		Pass@8				Pass@4		
Method	**Seed**	**AIME24**	**AIME25**	**AIME26**	**HMMT26**	**MATH500**	**Olympiad**	**GPQA-D**
Vanilla	42	26.67	33.33	26.67	21.21	87.00	53.41	46.46
Vanilla	1	26.67	33.33	23.33	21.21	86.20	53.12	47.47
Vanilla	2026	26.67	23.33	26.67	24.24	86.20	54.15	47.47
Vanilla	*Mean*	26.67	30.00	25.56	22.22	86.47	53.56	47.14
Direct Distillation	42	33.33	30.00	30.00	18.18	87.40	55.79	44.44
Direct Distillation	1	20.00	36.67	20.00	18.18	87.60	54.60	44.95
Direct Distillation	2026	23.33	30.00	26.67	18.18	88.80	56.97	42.42
Direct Distillation	*Mean*	25.56	32.22	25.56	18.18	87.93	55.79	43.94
Student RFT	42	13.33	40.00	23.33	24.24	88.20	51.63	50.00
Student RFT	1	20.00	40.00	16.67	21.21	85.60	53.12	48.99
Student RFT	2026	30.00	30.00	26.67	18.18	86.00	54.60	51.01
Student RFT	*Mean*	21.11	36.67	22.22	21.21	86.60	53.12	50.00
Teacher RFT	42	23.33	43.33	36.67	27.27	88.60	55.64	58.59
Teacher RFT	1	30.00	40.00	40.00	24.24	88.00	56.82	51.01
Teacher RFT	2026	23.33	40.00	33.33	24.24	88.20	54.90	53.54
Teacher RFT	*Mean*	25.56	41.11	36.67	25.25	88.27	55.79	54.38
IPD, S-P (1%)	42	23.33	40.00	33.33	24.24	90.40	56.38	53.03
IPD, S-P (1%)	1	33.33	46.67	26.67	27.27	90.20	59.20	54.04
IPD, S-P (1%)	2026	23.33	43.33	33.33	24.24	89.40	57.86	52.02
IPD, S-P (1%)	*Mean*	26.67	43.33	31.11	25.25	90.00	57.81	53.03
IPD, S-P (0.1%)	42	16.67	43.33	23.33	30.30	89.80	57.27	52.53
IPD, S-P (0.1%)	1	23.33	43.33	30.00	27.27	89.20	55.64	56.06
IPD, S-P (0.1%)	2026	26.67	36.67	30.00	27.27	89.60	55.64	52.02
IPD, S-P (0.1%)	*Mean*	22.22	41.11	27.78	28.28	89.53	56.18	53.54
IPD, S-P (0.01%)	42	33.33	43.33	33.33	24.24	89.00	57.27	52.53
IPD, S-P (0.01%)	1	30.00	36.67	26.67	24.24	89.00	55.64	51.01
IPD, S-P (0.01%)	2026	33.33	26.67	30.00	24.24	87.80	55.49	50.00
IPD, S-P (0.01%)	*Mean*	32.22	35.56	30.00	24.24	88.60	56.13	51.18
IPD, T-P (1%)	42	30.00	36.67	36.67	27.27	91.60	58.46	56.57
IPD, T-P (1%)	1	30.00	36.67	33.33	21.21	91.80	57.27	56.06
IPD, T-P (1%)	2026	30.00	36.67	30.00	30.30	89.20	58.61	58.59
IPD, T-P (1%)	*Mean*	30.00	36.67	33.33	26.26	90.87	58.11	57.07
IPD, T-P (0.1%)	42	26.67	33.33	43.33	27.27	91.00	58.61	58.08
IPD, T-P (0.1%)	1	33.33	33.33	30.00	27.27	90.20	57.27	62.12
IPD, T-P (0.1%)	2026	33.33	40.00	36.67	30.30	90.60	57.72	62.12
IPD, T-P (0.1%)	*Mean*	31.11	35.56	36.67	28.28	90.60	57.86	60.77
IPD, T-P (0.01%)	42	26.67	50.00	33.33	30.30	89.00	57.27	56.57
IPD, T-P (0.01%)	1	30.00	36.67	36.67	27.27	89.80	58.90	55.56
IPD, T-P (0.01%)	2026	40.00	36.67	30.00	27.27	88.00	59.35	56.06
IPD, T-P (0.01%)	*Mean*	32.22	41.11	33.33	28.28	88.93	58.51	56.06

Table 26: Pass@8 on AIME and HMMT and Pass@4 on the other benchmarks for Qwen3-1.7B/8B on s1K. Mean rows average coverage across three seeds.

		Pass@8				Pass@4		
Method	**Seed**	**AIME24**	**AIME25**	**AIME26**	**HMMT26**	**MATH500**	**Olympiad**	**GPQA-D**
Vanilla	42	60.00	50.00	56.67	39.39	95.20	67.51	64.14
Vanilla	1	56.67	56.67	66.67	39.39	94.60	68.99	64.65
Vanilla	2026	56.67	53.33	56.67	36.36	95.20	68.55	63.13
Vanilla	*Mean*	57.78	53.33	60.00	38.38	95.00	68.35	63.97
Direct Distillation	42	43.33	53.33	53.33	36.36	94.40	65.28	49.49
Direct Distillation	1	46.67	46.67	50.00	42.42	94.20	65.58	53.54
Direct Distillation	2026	50.00	43.33	53.33	33.33	95.60	65.43	52.53
Direct Distillation	*Mean*	46.67	47.78	52.22	37.37	94.73	65.43	51.85
Student RFT	42	56.67	53.33	63.33	42.42	94.40	67.36	62.63
Student RFT	1	56.67	53.33	60.00	45.45	95.00	68.69	58.08
Student RFT	2026	63.33	53.33	53.33	45.45	96.00	68.84	62.12
Student RFT	*Mean*	58.89	53.33	58.89	44.44	95.13	68.30	60.94
Teacher RFT	42	53.33	60.00	66.67	39.39	95.40	67.51	69.70
Teacher RFT	1	53.33	63.33	70.00	36.36	95.20	68.99	64.65
Teacher RFT	2026	63.33	60.00	73.33	45.45	96.60	67.80	68.69
Teacher RFT	*Mean*	56.67	61.11	70.00	40.40	95.73	68.10	67.68
IPD, S-P (1%)	42	56.67	50.00	63.33	42.42	95.20	68.69	63.13
IPD, S-P (1%)	1	56.67	56.67	66.67	39.39	94.80	67.51	62.63
IPD, S-P (1%)	2026	60.00	53.33	66.67	39.39	94.40	67.21	62.12
IPD, S-P (1%)	*Mean*	57.78	53.33	65.56	40.40	94.80	67.80	62.63
IPD, S-P (0.1%)	42	56.67	53.33	63.33	36.36	95.00	69.14	62.63
IPD, S-P (0.1%)	1	56.67	60.00	66.67	39.39	95.60	69.29	65.15
IPD, S-P (0.1%)	2026	53.33	53.33	70.00	42.42	95.40	69.44	63.64
IPD, S-P (0.1%)	*Mean*	55.56	55.56	66.67	39.39	95.33	69.29	63.80
IPD, S-P (0.01%)	42	60.00	56.67	63.33	42.42	94.60	69.44	59.09
IPD, S-P (0.01%)	1	60.00	53.33	63.33	39.39	95.20	70.47	70.71
IPD, S-P (0.01%)	2026	56.67	63.33	66.67	45.45	95.40	67.80	61.11
IPD, S-P (0.01%)	*Mean*	58.89	57.78	64.44	42.42	95.07	69.24	63.64
IPD, T-P (1%)	42	63.33	56.67	66.67	42.42	95.40	70.92	69.70
IPD, T-P (1%)	1	56.67	56.67	70.00	39.39	96.40	70.03	70.71
IPD, T-P (1%)	2026	53.33	63.33	60.00	42.42	95.20	68.55	68.18
IPD, T-P (1%)	*Mean*	57.78	58.89	65.56	41.41	95.67	69.83	69.53
IPD, T-P (0.1%)	42	60.00	63.33	70.00	45.45	95.60	69.14	68.69
IPD, T-P (0.1%)	1	60.00	66.67	70.00	36.36	95.80	70.62	69.19
IPD, T-P (0.1%)	2026	60.00	70.00	53.33	42.42	95.60	70.03	66.67
IPD, T-P (0.1%)	*Mean*	60.00	66.67	64.44	41.41	95.67	69.93	68.18
IPD, T-P (0.01%)	42	60.00	60.00	76.67	42.42	95.20	68.84	70.71
IPD, T-P (0.01%)	1	53.33	63.33	60.00	42.42	95.00	69.73	67.68
IPD, T-P (0.01%)	2026	53.33	56.67	66.67	45.45	96.20	69.29	67.17
IPD, T-P (0.01%)	*Mean*	55.56	60.00	67.78	43.43	95.47	69.29	68.52